



**Ministry of High Education
Culture and Science city at Oct 6,
The High Institute of Computer Science &
Information Systems**



Graduation Project

**Agri AI: Intelligent Plant Disease Recognition System deep
learning**

Supervised by:

Dr. Abeer Zidan

Assistant:

TA. Martina Safwat

Project No: F26_106

Academic Year :2025/2026



**Ministry of High Education
Culture and Science city at Oct 6,
The High Institute of Computer Science &
Information Systems**



Graduation Project

Agri AI: Intelligent Plant Disease Recognition System deep learning

Submitted by:

#	ID	Name
1	223815	زياد نادر عبد الله احمد
2	221123	حسام محمد مرزوق شحاته
3	220897	حمدي عرفه حمدي سيد
4	223455	علي عادل محمد محمد جوده
5	212892	زياد محمد عبد الرحمن عبد السميع
6	211930	درويش عبد الحميد درویش
7	220887	عاصم خميس احمد عبد الله

**A graduation project submitted in partial fulfillment of the
requirements for the degree of Bachelor of Computer Science and
Information System.**

Supervised by:

Dr. Abeer Zidan

Assistant:

TA. Martina Safwat

Project No : F26_106

Academic Year: 2025/2026

Acknowledgement

I would like to express my sincere gratitude to Culture and Science City and the Higher Institute for Computer Science and Information Systems for providing the academic environment and support that contributed to the completion of this project.

I would also like to extend my sincere appreciation to Dean **Prof. Sherif El-Atreby** for his leadership and continuous efforts in supporting students and fostering an environment that encourages academic excellence and innovation.

I would like to express my gratitude to all faculty members of the Higher Institute for Computer Science and Information Systems for their dedication, knowledge, and valuable contributions to our academic and professional development throughout our studies.

I would like to extend my deepest thanks and appreciation to **Dr. Abeer Zidan** for her continuous guidance, valuable feedback, and academic support throughout the development of this project. Her insights and supervision played a crucial role in shaping the project's outcomes.

I am also sincerely grateful to **Eng. Martina Safwat** for her technical support, encouragement, and constructive guidance, which greatly helped in improving the practical implementation of this work.

Finally, I would like to acknowledge all researchers and developers in the field of deep learning and convolutional neural networks, as well as the open-source community whose tools and libraries made this project possible.

Project Abstract

Plant diseases are among the most critical factors affecting agricultural productivity and crop quality, causing substantial economic losses and threatening global food security. Early identification of plant diseases is essential for effective treatment and prevention; however, conventional disease Recognition methods depend largely on visual inspection by agricultural experts, which is often time-consuming, expensive, and not always accessible, especially in rural areas.

This project presents an automated plant disease Recognition and classification system using **Convolutional Neural Networks (CNNs)**, aiming to provide a reliable and efficient alternative to traditional diagnosis techniques. The proposed system leverages deep learning to analyze images of plant leaves and accurately identify disease types at early stages. A comprehensive dataset of labeled leaf images representing healthy plants and multiple disease categories is utilized to train the model.

Several preprocessing steps are applied to enhance model performance, including image resizing, normalization, and data augmentation techniques such as rotation and flipping to improve robustness and reduce overfitting. The CNN architecture is designed to automatically extract high-level features from raw images, eliminating the need for manual feature engineering. The model is trained and evaluated using standard evaluation metrics such as accuracy, loss, and validation performance.

Experimental results demonstrate that the developed CNN model achieves high accuracy in classifying plant diseases, confirming the effectiveness of deep learning approaches in agricultural image analysis. The system shows

Table of Contents

Acknowledgment.....	3
Project Abstract.....	4
Chapter1: Introduction	10
1.1 Overview	11
1.2 Problem Statement	12
1.3 Motivation	13
1.4 Objective	14
1.5 Aim	14
1.6 Scope	14
1.7 General Constraints	15
1.8 Organization of the Project.....	15
Chapter 2 Background and Previous work	16
2.1 Background.....	17
2.2 Previous work.....	18
Chapter 3 Planning and Analysis	20
3.1 Planning.....	21
3.1.1 Feasibility study and estimated cost.....	22
3.1.2 Gantt chart.....	23
3.2 Analysis and limitations of existing system.....	24
3.3 Need for new system.....	25
3.4 Analysis of new system.....	26
3.4.1 Introduction	26
3.4.2 Overall Description	27
3.4.3 Specific Requirements	29
3.4.4.1 Functional Requirements.....	29

3.4.4.2 Non- Functional Requirements	30
3.5 External Interface Requirements.....	31
Chapter 4: Design	32
4.1 Design and Implementation Constraints.....	33
4.2 Assumptions and dependencies.....	36
4.3 Risks and risk management.....	37
4.4 User Experience (UX)	39
4.4.1 Problem statement.....	40
4.4.2 Persona.....	41
4.4.3 Empathy map.....	42
4.4.4 User journey map.....	43
4.4.5 Information Architecture (IA).....	44
4.4.6 User Flow.....	45
4.4.7 Wireframes.....	46
4.4.7.1 Low-fidelity wireframes.....	47
4.4.7.2 High-fidelity wireframes.....	49
4.5 User Interface (UI).....	51
4.6 Design of database ERD.....	53
4.7 Class diagram.....	53
4.8 Use case diagram.....	54
4.9 Activity diagram.....	56
4.10 Sequence diagram.....	57
4.11 Context diagram.....	58

Chapter 5 Implementation.....	59
5.1 Software Architecture	60
5.2 User Interface.....	63
5.3 Results and Discussion.....	65
Chapter 6: Testing.....	68
6.1 Unit Testing.....	69
6.2 Integration Testing.....	70
Conclusion.....	72
Future work	74
References	76
Appendices	77
ملخص مشروع نظام التعرف على أمراض النبات.....	78

Table of Figures

Figure 3.1 (Gantt Chart)	23
Figure 4.1 (Problem statement)	40
Figure 4.2 (Persona)	41
Figure 4.3 (Empathy Map)	42
Figure 4.4 (User Journey Map)	43
Figure 4.5 (Information Architecture (IA))	44
Figure 4.6 (User Flow)	45
Figure 4.7 (Low-Fidelity Wireframes-Homepage)	48
Figure 4.8 (Low-Fidelity Wireframes- Upload Image page)	48
Figure 4.9 (Low-Fidelity Wireframes- Diagnosis Result page.....	48
Figure 4.10 (High-Fidelity Wireframes- Homepage)	50
Figure 4.11 (High-Fidelity Wireframes- Upload Image page)	50
Figure 4.12 (High-Fidelity Wireframes- Diagnosis Result page)	50
Figure 4.13 (Homepage UI)	52
Figure 4.14 (Upload Image page UI).....	52
Figure 4.15 (Diagnosis Result page UI)	52
Figure 4.16 (Class Diagram)	53
Figure 4.17 (Use-case Diagram)	55

Figure 4.18 (Sub Use-case Diagram)	55
Figure 4.19 (Activity Diagram)	56
Figure 4.20 (Sequence Diagram)	57
Figure 4.21 (Context Diagram)	58
Figure 5.1: (Image Preprocessing)	60
Figure 5.2: (CNN Architecture)	61
Figure 5.3: (Model Deployment and Prediction Interface Implementation)	64
Figure 5.4: (Training and Validation Accuracy)	65
Figure 5.5: (Training and Validation Loss)	66
Figure 5.6: (Treatment Recommendation System)	67
Figure 6.1: (System Integration Workflow)	70
Figure 6.2: (Integrated System Output)	71

Chapter 1

Introduction

1.1 Overview

Agriculture is considered one of the most essential sectors worldwide, as it plays a fundamental role in providing food, raw materials, and employment opportunities for a large portion of the global population. The agricultural sector significantly contributes to economic stability and food security, especially in developing countries. However, agricultural productivity faces many challenges, among which plant diseases are one of the most serious and damaging factors.



Plant diseases are a major threat to agricultural productivity, causing significant crop losses if not detected early. Traditional disease diagnosis relies on manual inspection and expert knowledge, which can be time-consuming, costly, and unavailable in many rural areas.

Recent advances in artificial intelligence, particularly deep learning, have provided effective solutions for automated disease detection. Convolutional Neural Networks (CNNs) have demonstrated high performance in image classification and recognition tasks.

1.2 Problem Statement

Plant disease Recognition using traditional manual methods presents several challenges that limit its effectiveness and accessibility. Farmers often rely on visual inspection and personal experience to identify plant diseases, which can lead to incorrect diagnoses due to similarities between disease symptoms or lack of sufficient expertise. In many cases, early symptoms of diseases are subtle and difficult to detect with the naked eye, causing delayed intervention.

Furthermore, access to agricultural experts is limited in many regions, especially in rural and developing areas. This lack of expert support results in late disease identification, which allows diseases to spread rapidly across crops. Consequently, farmers may resort to excessive or inappropriate use of pesticides, leading to increased costs, environmental pollution, and potential health risks.

Existing challenges can be summarized as follows:

- Manual disease Recognition is time-consuming and prone to human error.
- Accurate diagnosis requires expert knowledge that is not always available.
- Delayed Recognition leads to significant crop losses and reduced productivity.

1.3 Motivation

The motivation for this project stems from the increasing demand for smart agricultural solutions that leverage modern technologies to improve efficiency and productivity. With the rapid growth of the global population, the need for higher agricultural output has become more urgent than ever. Reducing crop losses caused by plant diseases is essential to meet this demand.

The success of deep learning models, particularly CNNs, in image classification tasks has inspired their application in agriculture. CNNs are capable of automatically extracting relevant features from images without manual feature engineering, making them suitable for analyzing complex visual patterns found in diseased plant leaves.

Another major motivating factor is the widespread availability of digital imaging devices such as smartphones and cameras. This availability makes it feasible to capture plant images easily and use automated systems for disease diagnosis. The development of an AI-based plant disease Recognition system can empower farmers with limited technical knowledge by providing them with a simple yet effective diagnostic tool.

1.4 Objective

The main objectives of this project are as follows:

- To study and analyze plant diseases and their visual symptoms on leaves.
- To design and implement a Convolutional Neural Network for plant disease classification.
- To preprocess plant leaf images to enhance model accuracy and performance.
- To train and evaluate the model using labeled datasets of healthy and diseased leaves.

1.5 Aim

The main goal of this project is to create an automated and intelligent plant disease recognition system that accurately classifies diseases from leaf images. It helps farmers and agricultural experts by enabling early detection, reducing crop losses, and improving agricultural productivity.

1.6 Scope

The scope of this project includes recognizing and classifying plant diseases using leaf images only. It focuses on applying supervised deep learning techniques, particularly CNNs, along with image preprocessing methods such as resizing, normalization, and augmentation. The project also includes evaluating the model's performance using accuracy and loss metrics.

1.7 General Constraints

Despite the advantages of the proposed system, certain constraints are identified:

- The accuracy of the model depends on the quality and diversity of the dataset.
- Variations in lighting conditions, background noise, and image resolution may affect results.
- High computational resources are required for training deep learning models.
- The system may not perform well on diseases or plant species not included in the training data.

1.8 Organization of the Project

This project is structured into the following chapters:

- **Chapter 1:** Introduction, problem statement, objectives, scope, and constraints.
- **Chapter 2 :**Background and Previous work
- **Chapter 3:** Planning And Analysis
- **Chapter 4:** Design
- **Chapter 5:** Implementation.
- **Chapter 6:** Testing

Chapter 2

Background and Previous work

2.1 Background

Plant diseases significantly reduce agricultural productivity and threaten global food security due to yield losses caused by fungi, bacteria, viruses, and nematodes. These diseases affect different parts of plants and often appear as visible symptoms such as spots, discoloration, wilting, and deformation. Traditionally, plant disease detection depends on manual inspection by experts, which can be accurate but is limited by availability, subjectivity, and difficulty in identifying early-stage infections, leading to delays and increased spread of disease.

To overcome these limitations, automated systems based on computer vision and machine learning have been developed to detect and classify plant diseases from images. These systems provide faster, more consistent, and scalable solutions compared to manual methods. Machine learning techniques map image features to disease categories, but they often require manual feature extraction such as color, texture, and shape, which can be time-consuming and dependent on expert knowledge.

Deep learning, especially Convolutional Neural Networks (CNNs), has greatly improved image classification by automatically learning features directly from raw images. CNNs detect patterns like edges, textures, and complex structures that help distinguish between healthy and diseased plants with high accuracy. The use of CNNs in plant disease recognition offers advantages such as automated feature extraction, improved accuracy, better generalization, and scalability across different crops and disease types.

This project is based on CNN technology to develop an efficient and reliable plant disease recognition system that combines image processing and artificial intelligence to support smart and modern agriculture.

2.2 Previous Work

Over the past decade, considerable research efforts have been made to develop automated plant disease Recognition systems. These efforts span multiple methodologies, datasets, model architectures, and performance evaluations. In this section, we present some of the most relevant and influential studies in the field.

Traditional Machine Learning Approaches

Early research in plant disease detection relied on traditional machine learning methods such as SVM, KNN, and Random Forests after extracting manual image features. These approaches used features like color, texture, and LBP from leaf images and achieved acceptable accuracy in some cases. However, their performance depended heavily on handcrafted features and extensive preprocessing, which limited their ability to generalize to different disease conditions.

Deep Learning Approaches

With the rise of deep learning, researchers started using CNN models to improve plant disease recognition. A key study by Mohanty et al. (2016) used AlexNet and GoogleNet on the PlantVillage dataset and achieved very high accuracy, showing the strength of deep learning in this field. Later studies, such as Too et al. (2019), compared different CNN architectures like VGGNet, ResNet, and Inception, and found that deeper models like ResNet

Hybrid and Transfer Learning Methods

Researchers have also used hybrid methods that combine deep learning with traditional techniques or apply transfer learning using pre-trained models like those trained on ImageNet. Ferentinos (2018) showed that transfer learning with CNNs can achieve high accuracy and reduce training time even with limited labeled data, proving that general image knowledge can be adapted for plant disease recognition.

Other studies also used image segmentation to focus on infected areas of leaves before classification. Although this can improve accuracy in some cases, it adds extra preprocessing steps and increases system complexity.

Limitations of Previous Work

Despite the progress and high accuracies reported in research studies, several limitations persist:

Many models are trained and evaluated on controlled datasets with uniform backgrounds, which do not reflect real-world field conditions.

Performance may degrade when tested on images captured under varying lighting, angles, and backgrounds.

Some approaches require high computational resources.

Chapter 3

Planning And

Analysis

3.1 Planning

The planning phase represents the foundation of the project development process. It focuses on identifying the required steps, resources, and timeline needed to build the plant disease Recognition system using Convolutional Neural Networks (CNN). This phase includes determining the feasibility of the project, estimating the cost, identifying potential risks, organizing the workflow, and designing a clear schedule for project execution. Proper planning ensures that every stage is aligned with the project objectives and that all tasks are carried out efficiently within a reasonable time frame and available resources.

The planning phase also includes defining team responsibilities, understanding the required dataset, selecting development tools, and specifying the evaluation criteria that will be used to measure the success of the system. This organized structure contributes to reducing errors, avoiding delays, and ensuring the successful completion of the project.

3.1.1 Feasibility Study and Estimated Cost

The feasibility study aims to determine whether the project can be executed effectively based on technical, economic, and operational factors. It confirms if the proposed system is achievable with the available tools, time, and team capabilities.

A. Technical Feasibility

- The project is technically feasible because it relies on Python, TensorFlow/Keras, CNN models, and publicly available datasets such as PlantVillage.
- Deep learning tools like Google Colab or Kaggle Notebooks can be used for training without the need for purchasing high-end hardware.
- Image preprocessing and dataset handling are manageable with libraries such as OpenCV, NumPy, and Pandas.

Conclusion: The project is technically achievable with current resources.

B. Estimated Cost

Item / Resource	Description / Usage	Estimated Cost
Laptop/PC (Existing)	System development & testing	0 L.E (Available)
Google Colab	Model training environment	0 L.E (Free Tier)

Item / Resource	Description / Usage	Estimated Cost
Python Libraries & Frameworks	TensorFlow, Keras, NumPy, OpenCV, etc.	0 L.E (Open-source)
Dataset	PlantVillage Dataset	0 L.E (Free)
Internet Usage / Power Consumption	Research & model training	~200 L.E
Documentation & Printing	Final report & presentation	~150 L.E

3.1.2 Gantt Chart (Project Timeline)



Figure 3.1(Gantt Chart)

3.2 Analysis and Limitations of Existing System

Current plant disease Recognition systems vary from manual visual inspection by experts to semi-automated software solutions. Despite their widespread use, existing systems present several limitations:

3.2.1 Manual Inspection

Traditional methods rely on expert knowledge to identify diseases by visual symptoms. Limitations include:

1. Time-consuming: Inspecting large fields manually requires substantial effort.
2. Inaccuracy: Human observation is prone to errors, especially in the early stages of disease.
3. Limited Access: Many rural areas lack sufficient agricultural specialists.

Semi-Automated and Machine Learning Systems

Earlier automated systems use image processing and classical machine learning models (e.g., SVM, KNN, Random Forest) to classify diseases.

Limitations include:

1. Dependence on handcrafted features (color, texture, shape), which may not capture subtle disease characteristics.
2. Low generalization to different lighting conditions, backgrounds, or leaf orientations.
3. Performance degradation on complex or noisy datasets.

3.3 Need for New System

The proposed system aims to overcome the shortcomings of existing solutions by providing an automated, scalable, and accurate plant disease Recognition platform. The main reasons for developing a new system include:

1. **Enhanced Accuracy and Reliability**

Deep learning-based models, particularly CNNs, can automatically extract high-level features from leaf images, improving the Recognition of subtle disease symptoms.

2. **Time and Resource Efficiency**

The automated system reduces the need for manual inspection, allowing farmers and experts to diagnose diseases quickly across large areas.

3. **Wider Accessibility**

By integrating the system into a web-based or mobile application, farmers in remote or rural areas can access accurate disease diagnostics without relying on experts.

4. **Adaptability and Scalability**

The system can be trained to include new plant species and disease types as more data becomes available, making it scalable for various crops and regions.

5. **Support for Sustainable Agriculture**

Accurate early Recognition reduces unnecessary pesticide use,

3.4 Analysis of New System

3.4.1 Introduction

Purpose

This document provides a detailed Software Requirements Specification (SRS) for the **Intelligent Pest Plants Recognition System**. The purpose of this system is to automatically detect plant pests and diseases using intelligent image processing and machine learning techniques. This document is intended for software developers, system designers, project managers, and stakeholders involved in the development and deployment of the system.

Scope

The Intelligent Pest Plants Recognition System helps identify plant pests and diseases from uploaded images using trained models, providing recognition results and recommendation Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification
- **AI:** Artificial Intelligence
- **ML:** Machine Learning
- **CNN:** Convolutional Neural Network
- **User:** Any individual using the system (farmer, researcher, admin)

3.4.2 Overall Description

Product Perspective

The Intelligent Pest Plants Recognition System is a standalone web-based and/or mobile-based application. It integrates image acquisition, preprocessing, machine learning-based pest detection, and result visualization modules.

Product Functions

- Upload plant images for analysis
- Detect pests and plant diseases using AI models
- Display Recognition results with confidence levels
- Provide pest information and basic treatment suggestions
- Store analysis history
- Manage users and system settings (admin)

User Classes and Characteristics

- **Farmers:** Basic users with minimal technical knowledge
- **Agricultural Experts:** Users with domain expertise
- **System Administrator:** Responsible for system maintenance and management

Operating Environment

- Client-side: Web browser or Android/iOS mobile application

- Server-side: Cloud or local server
- Database: SQL or NoSQL database
- AI Frameworks: TensorFlow / PyTorch

Design and Implementation Constraints

- Availability of labeled pest datasets
- Internet connectivity for cloud-based processing
- Hardware limitations of user devices

Assumptions and Dependencies

- Users provide clear images of plants
- AI models are pre-trained and periodically updated
- System accuracy depends on dataset quality

3.4.3 Specific Requirements

3.4.3.1 Functional Requirements

User Registration and Login

- The system shall allow users to register using email and password.
- The system shall authenticate users securely.

Image Upload

- The system shall allow users to upload plant images in common formats (JPEG, PNG).
- The system shall validate image size and format.

Pest Detection

- The system shall analyze uploaded images using machine learning models.
- The system shall identify pests or diseases present in the image.
- The system shall return Recognition results with confidence percentages.

Result Display

- The system shall display the detected pest name.
- The system shall give a brief description and possible treatment methods.

3.4.3.2 Non-Functional Requirements

Performance Requirements

- The system shall process each image within an acceptable response time (less than 5 seconds).

Security Requirements

- The system shall ensure secure authentication and authorization.
- The system shall protect user data and uploaded images.

Usability Requirements

- The system shall provide a simple and intuitive user interface.
- The system shall support multiple languages.

Reliability Requirements

- The system shall be available 99% of the time.
- The system shall handle system failures gracefully.

Scalability Requirements

- The system shall support increasing numbers of users and image uploads.

3.4.4 External Interface Requirements

User Interface

- Responsive graphical user interface
- Simple upload and result display screens

Hardware Interfaces

- Camera (mobile devices)
- Server GPU (optional for faster processing)

Software Interfaces

- AI libraries (TensorFlow / PyTorch)
- Database management system

Chapter 4

Design

4.1 Design and Implementation Constraints

The design and implementation of the proposed Plant Disease Recognition System are subject to several technical, operational, and environmental constraints that must be considered to ensure successful development and deployment.

1. Model and Algorithm Constraints

- The system must rely on a Convolutional Neural Network (CNN) architecture for disease Recognition and classification.
- The CNN model architecture and training process should be documented and remain consistent unless retraining or improvement is explicitly planned.
- The model performance is constrained by the quality, size, and diversity of the training dataset.

2. Dataset Constraints

- The system will use pre-collected and publicly available datasets (e.g., PlantVillage).
- The dataset may not cover all plant species or disease variations found in real-field environments.
- Any imbalance in the dataset (unequal number of samples per class) may affect classification accuracy.

3. Image Input Constraints

- Uploaded images must be in supported formats such as JPEG or PNG.
- Image size and resolution must fall within predefined limits (e.g., maximum file size and minimum resolution) to ensure proper processing.
- Poor-quality images (blurred, low lighting, occluded leaves) may lead to reduced Recognition accuracy.

4. Hardware and Computational Constraints

- Model training may require significant computational resources, including GPU acceleration for faster convergence.
- Inference (prediction) should be optimized to run efficiently on standard CPU-based systems without requiring specialized hardware.
- The system must operate within the limitations of available memory and processing power, especially if deployed on low-cost devices.

5. Software and Platform Constraints

- The system uses open-source technologies such as Python, TensorFlow/Keras, OpenCV, and Flask/FastAPI.
- It is compatible with common operating systems and modern web browsers.

6. Performance Constraints

- The system should provide diagnosis results within an acceptable response time (e.g., ≤ 5 seconds per image on CPU).
- The system must handle single-user and limited multi-user requests without significant performance degradation.

7. Security and Privacy Constraints

- The system must not store or expose sensitive personal user data.
- Uploaded images and diagnosis results should be handled securely, especially if the system is deployed online.
- Unauthorized access to the system or its backend services must be prevented.

8. Deployment Constraints

- If deployed as a web-based application, the system must run on a standard web server without requiring specialized infrastructure.
- Internet connectivity is required for online deployment; offline functionality may be limited to local installations.

9. Maintenance and Extensibility Constraints

- The system design must be modular to allow future updates, such as adding new diseases, plants, or improving the CNN model.
- Any extension of the system may require additional data, retraining, and validation of the model.

4.2 Assumptions and Dependencies

The design and implementation of the Plant Disease Recognition System are based on a set of assumptions and dependencies that affect system functionality and performance.

Assumptions

- Users are assumed to provide clear and properly captured images of plant leaves.
- The uploaded images are expected to show visible disease symptoms in case of infection.
- The dataset used for training is assumed to be representative of the target plant species and diseases.
- Users have access to a device capable of capturing or uploading images.
- For web-based deployment, users are assumed to have a stable internet connection.

Dependencies

- The system depends on a trained Convolutional Neural Network (CNN) model for disease Recognition and classification.
- The implementation relies on external libraries and frameworks such as Python, TensorFlow/Keras, and OpenCV.

4.3. Risk and Risk Management

This project involves several potential risks that may affect system performance, development timeline, or usability. Risk management is applied to identify these risks early and define strategies to reduce their impact.

Risk	Description	Impact	Mitigation Strategy
Low Image Quality	Users may upload blurred or poorly lit images	Reduced diagnosis accuracy	Input validation, user instructions, image preprocessing
Dataset Limitations	The system relies on a pre-collected dataset of plant leaf images, often captured in controlled environments. These “clean” images may	Limited generalization, reduced performance in field conditions	Data augmentation, testing with real-field images, future dataset expansion

Risk	Description	Impact	Mitigation Strategy
	not represent real-world variability (lighting, backgrounds, leaf orientations, partial occlusions). Also, some plant species or rare diseases may not be available in the dataset		
Overfitting	Models may perform well on training data but poorly on new images	Poor real-world performance	Regularization techniques, validation, early stopping

Risk	Description	Impact	Mitigation Strategy
High Training Time	CNN training may require long computation time	Development delays	Use pre-trained models, cloud platforms (Colab/Kaggle)

4.4 User Experience (UX)

User Experience (UX) in the Plant Disease Recognition System focuses on providing an intuitive and efficient interaction for users. The system allows users to identify plant diseases easily by uploading a plant leaf image and receiving instant diagnostic results. The design minimizes user effort by reducing the interaction steps and presenting clear, direct functionality.

The interface emphasizes simplicity and visual clarity through a clean layout and nature-inspired colors. Key elements such as the image upload area and diagnosis results are clearly highlighted, ensuring ease of use. Overall, the UX design aims

4.4.1 Problem Statement

This is a Problem Statement for a plant disease recognition system project. The text describes the current challenges faced by farmers and non-specialized users in identifying plant diseases accurately. Traditional methods rely heavily on visual inspection and expert knowledge, which are often time-consuming, subjective, and not always available. In addition, many existing digital solutions suffer from complex user interfaces, making them difficult to use efficiently.

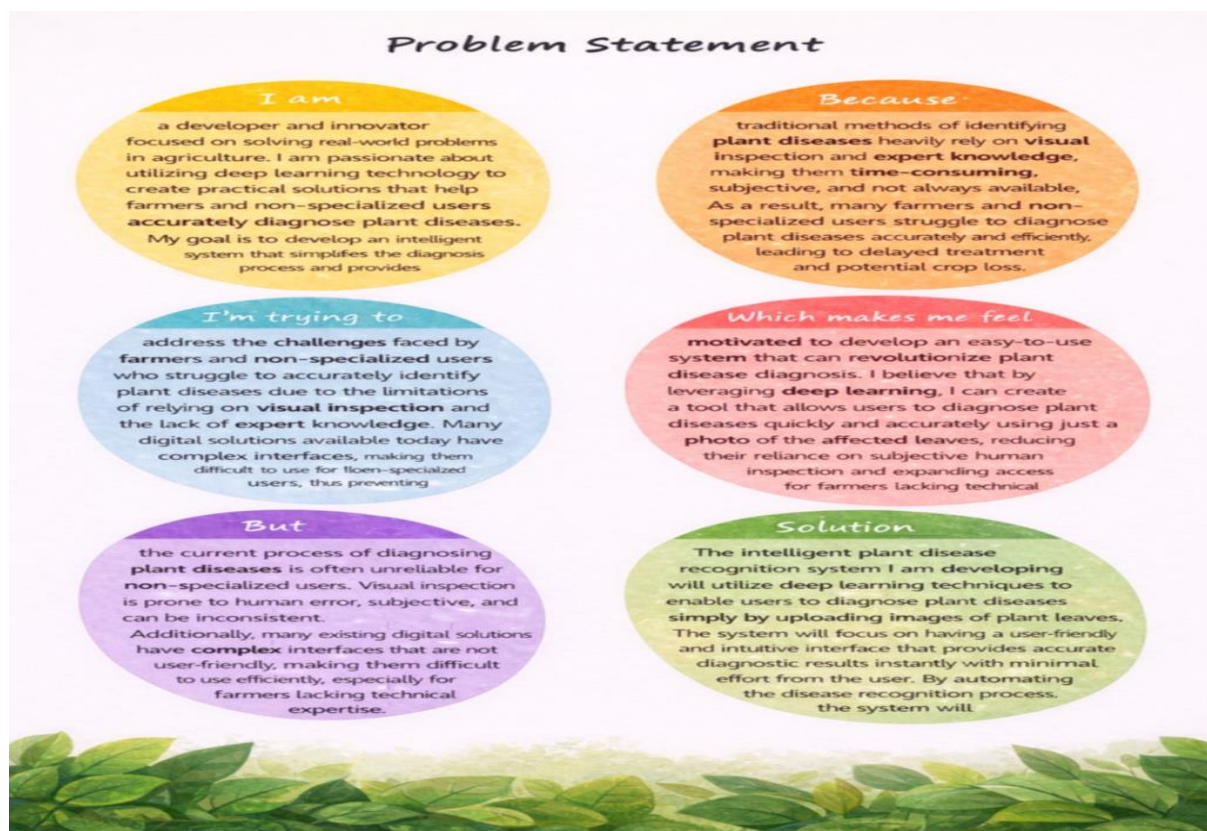


Figure 4.1 (Problem statement)

4.4.2 Persona

This is a User Persona for a farmer or non-specialized user who faces difficulties in identifying plant diseases accurately. The user relies on traditional methods and lacks expert knowledge, which leads to delays and uncertainty in diagnosis.

The user's goal is to obtain quick and accurate results through a simple and clear interface. The system is designed to minimize complexity by allowing users to upload a plant leaf image and receive immediate diagnostic feedback. This persona supports the design of a user-friendly interface that meets the needs of users with minimal technical experience.

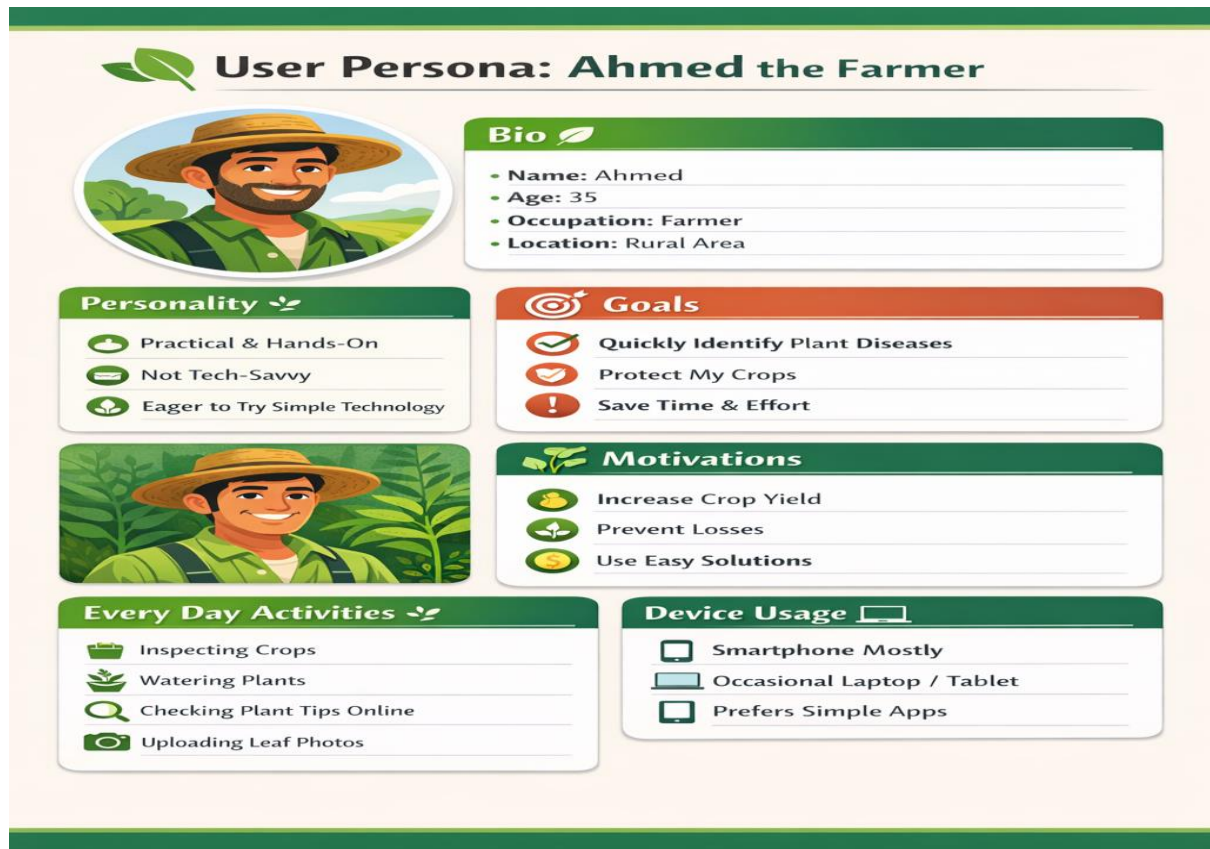


Figure 4.2 (Persona)

4.4.3 Empathy Map

This is an Empathy Map for a farmer using the Plant Disease Recognition System. It outlines the user's thoughts, feelings, actions, and concerns regarding plant disease diagnosis. The map highlights frustrations with traditional manual methods, the need for simple and immediate diagnostic tools, and

the desire to protect crops efficiently. By understanding the user's emotions and behaviors, the system is designed to provide clear, intuitive, and actionable feedback, enhancing the overall user experience.

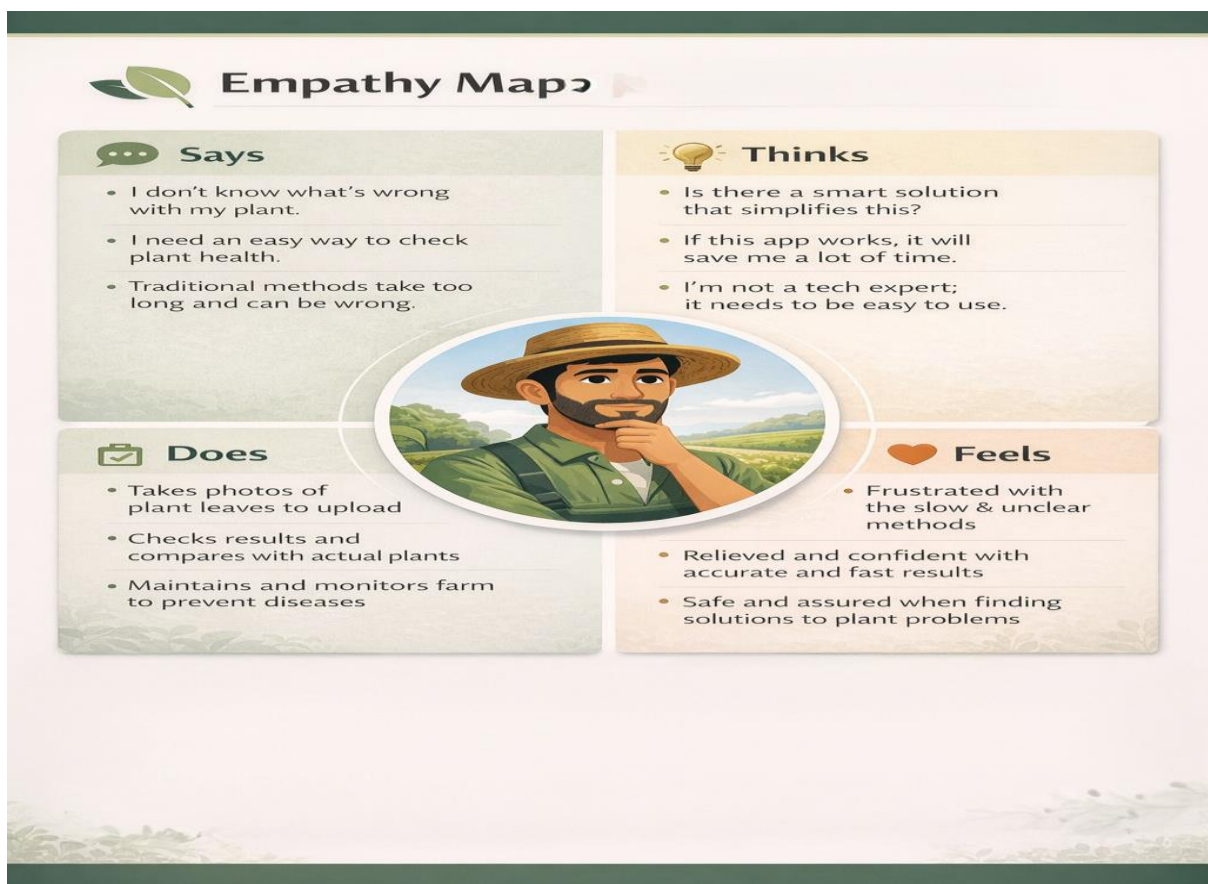


Figure 4.3 (Empathy Map)

4.4.4 User Journey Map

This is a User Journey Map for a farmer using the Plant Disease Recognition System. It outlines the steps the user goes through from planning which plants to check, capturing images, analyzing results, making treatment decisions, to searching for additional solutions. The map highlights challenges such as uncertainty in diagnosing plant diseases, the need for accurate and fast feedback, and the emotional responses at each stage. By understanding the user's journey, the system is designed to provide clear guidance, reduce stress, and support informed decision-making for effective crop management.

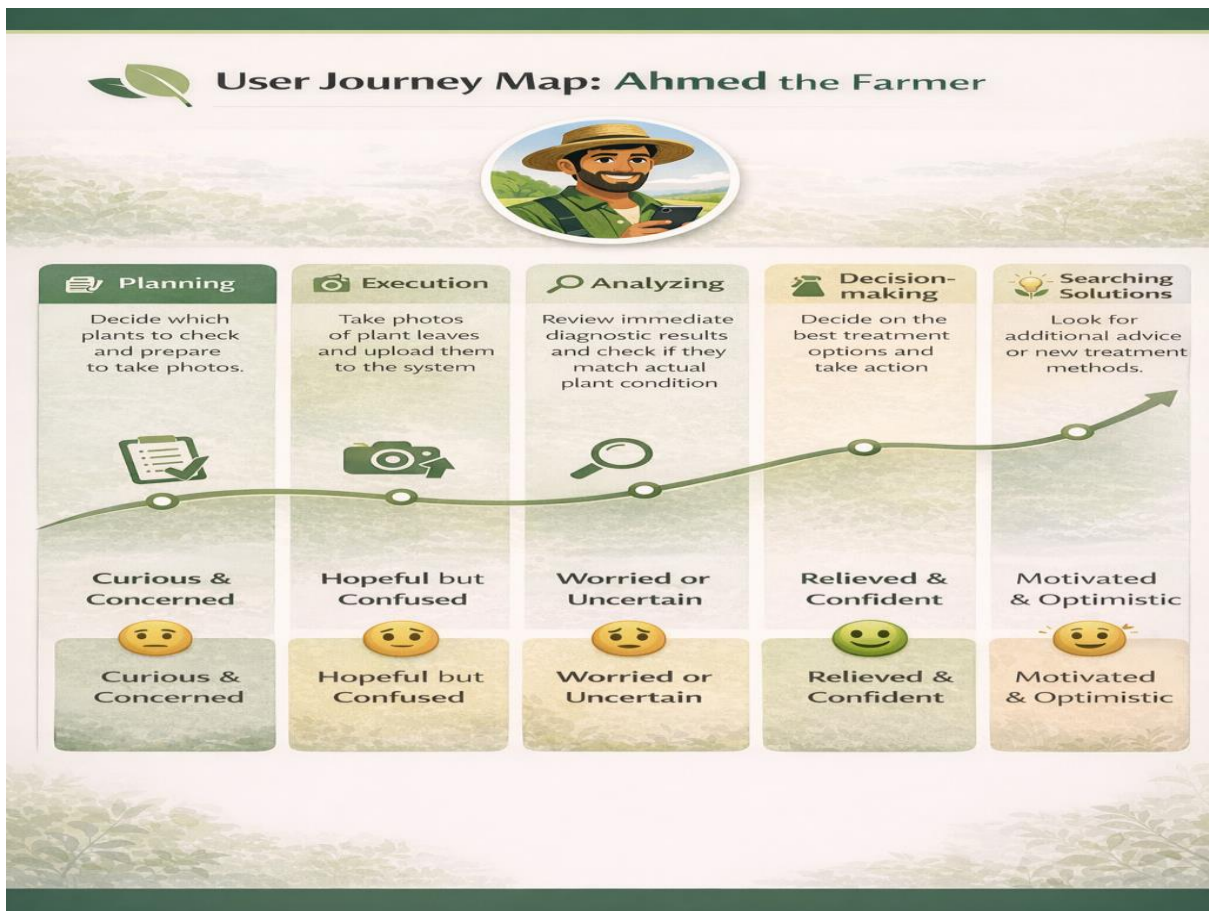


Figure 4.4 (User Journey Map)

4.4.5 Information Architecture (IA)

This is the Information Architecture of the Plant Disease Recognition System. It starts from the **Home Page**, leading to various modules such as **Login/Register**, **Dashboard**, **Upload Image**, **Diagnosis History**, **Plant Species Database**, and **User Settings**.

Each section contains specific functionalities, for example:

- **Dashboard** provides an overview of recent diagnoses and system status.
- **Upload Image** allows users to submit plant leaf images for instant disease detection.
- **Diagnosis History** stores previous results for easy reference and monitoring.
- **Plant Species Database** provides information about different plants and common diseases.
- **User Settings** manage account information and preferences.

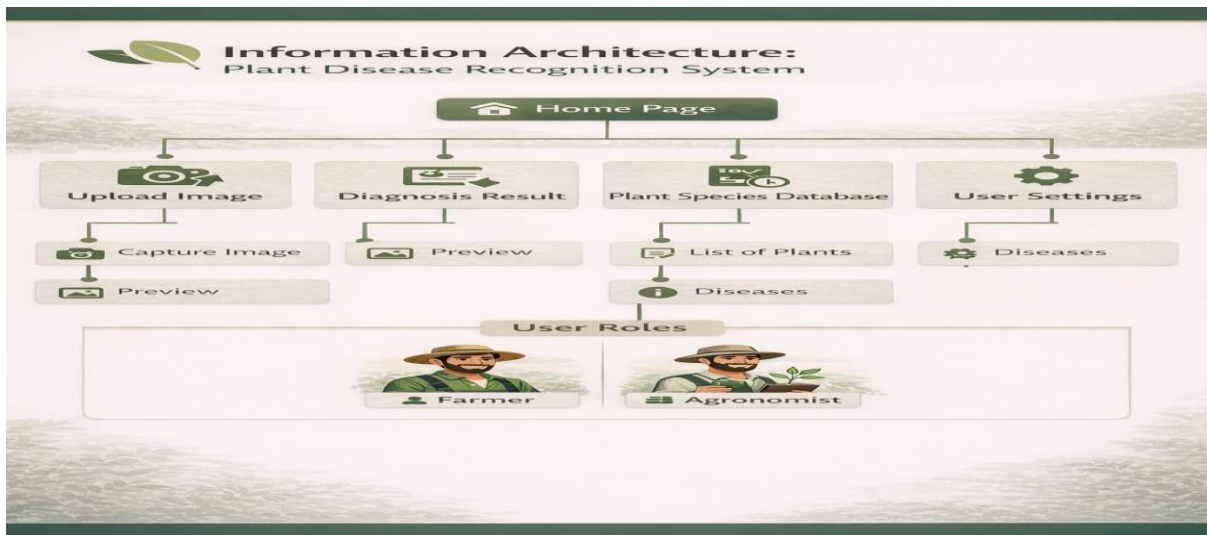


Figure 4.5 (Information Architecture (IA))

4.4.6 User Flow

The user flow begins when the user opens the web application and uploads an image of a plant leaf. The system then processes the image using the trained deep learning model and displays the diagnosis result, including the detected disease and confidence score. Users can also browse the plant species database to explore different plants and their associated diseases.

Flow Steps:

1. Open the Plant Disease Recognition System website.
2. Upload an image of the plant leaf.
3. System predicts disease and displays results:
 - Disease Name
 - Confidence
 - Related Plant Information

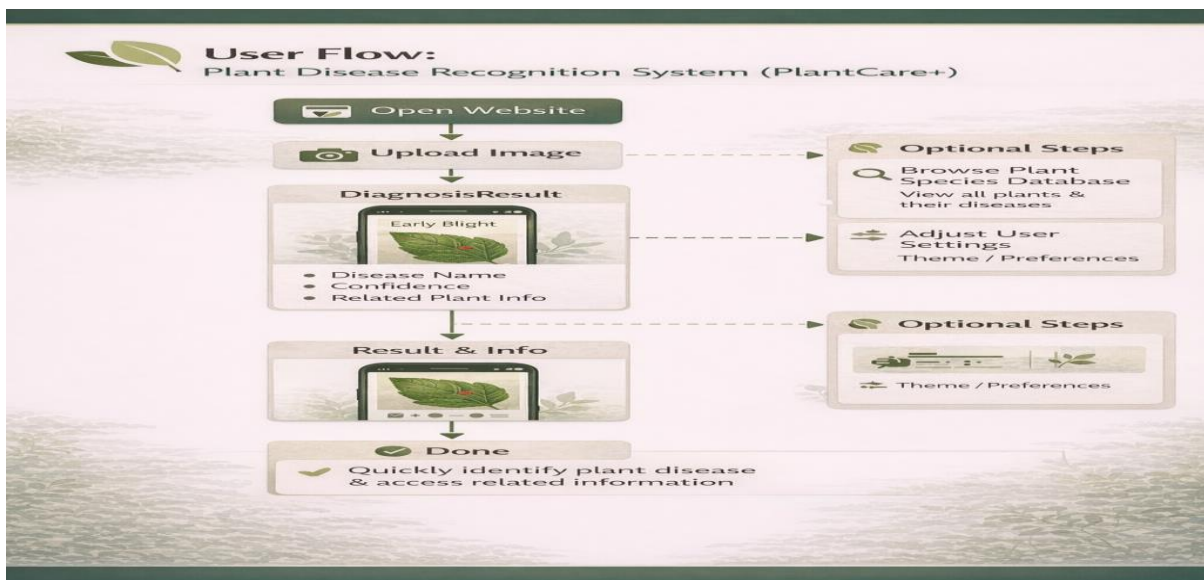


Figure 4.6 (User Flow)

4.4.7 Wireframes

Wireframes for the Plant Disease Recognition System serve as visual blueprints of the application's structure and layout. They focus on functionality and placement of content, without emphasizing colors, fonts, or visual styling.

The main wireframes include:

1. Home Page Wireframe

- **Header:** Project title and navigation links.
- **Main Section:** Hero section with a brief description and call-to-action to upload an image.
- **Footer:** Basic footer with copyright or project info.

2. Upload Image Wireframe

- **Upload Button:** Allows users to select a plant leaf image.
- **Preview Area:** Displays the uploaded image.
- **Instruction Text:** Guides the user on acceptable image formats and size.

3. Diagnosis Result Wireframe

- **Result Card:** Displays predicted disease name, confidence score, and related plant information.

4.4.7.1 Low-Fidelity Wireframes

Low-fidelity (Lo-Fi) wireframes are simple, sketch-like representations of the Plant Disease Recognition System layout. They focus on **structure, functionality, and user flow** rather than visual design details. Typically, they use **boxes, lines, and placeholders** for text, buttons, and images.

How They Are Created for this Project:

1. **Hand-drawn sketches** – Quickly drafted on paper or whiteboards to outline the main screens:
 - **Home Page** with Hero section and Upload Image button.
 - **Upload Image Page** showing the image preview area.
 - **Diagnosis Result Page** displaying disease name and confidence.
2. **Basic grayscale design** – Avoids colors, fonts, or real images, focusing only on **layout, navigation, and result placement**.

Lo-Fi wireframes are ideal for brainstorming, gathering initial feedback, and making quick changes to the **Plant Disease Recognition System interface** before moving to high-fidelity designs.

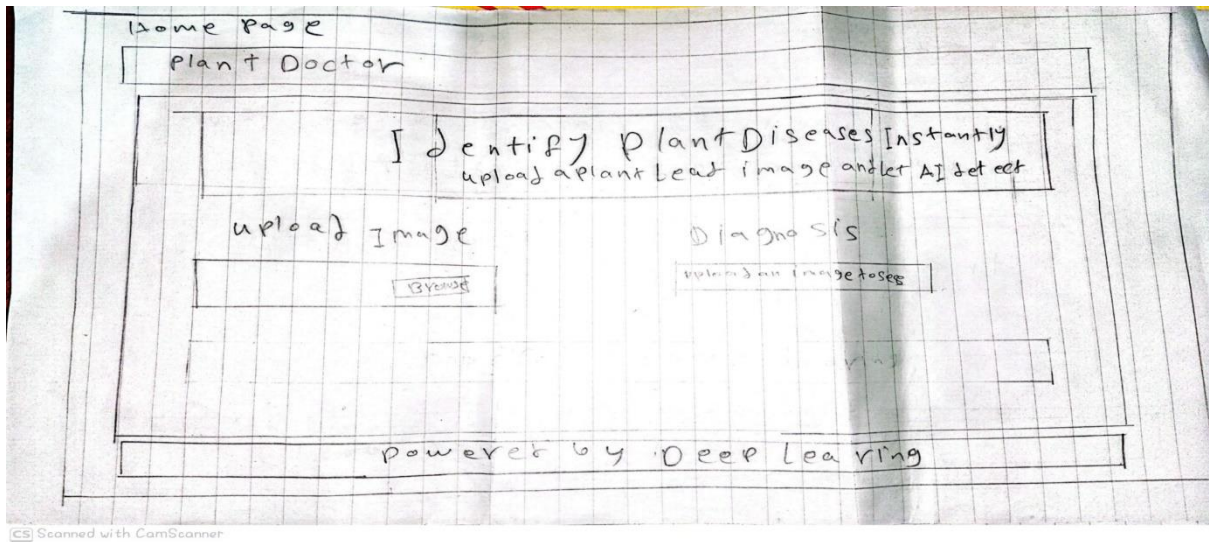


Figure 4.7(Low-Fidelity Wireframes-Homepage)

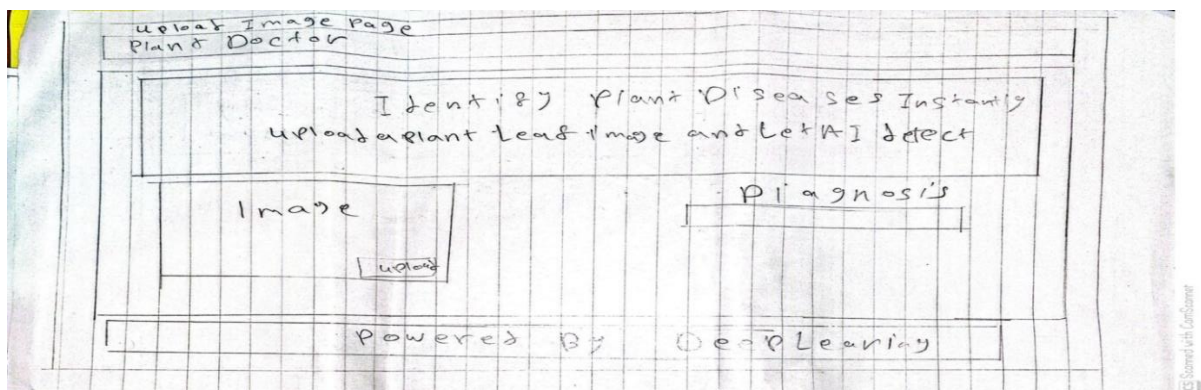


Figure 4.8 (Low-Fidelity Wireframes- Upload Image page)

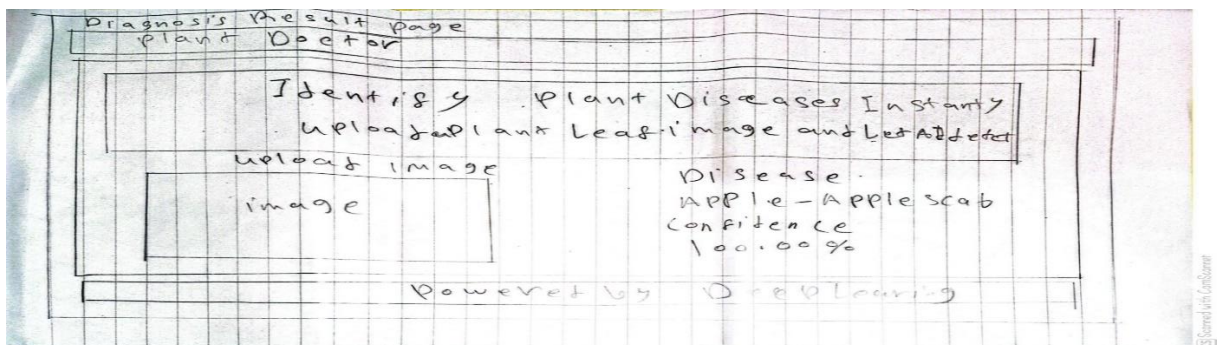


Figure 4.9 (Low-Fidelity Wireframes- Diagnosis Result page)

4.4.7.2 High-Fidelity Wireframes

High-fidelity (Hi-Fi) wireframes for the Plant Disease Recognition System are detailed, visually refined, and closely resemble the final application. They use **real UI elements, images, text, and buttons**, giving an accurate representation of how the user will interact with the system.

How They Are Created for this Project:

1. **Using design software** – Tools like Figma, Adobe XD, or Sketch can be used to recreate the Stream lit interface of the Plant Disease Recognition System.
2. **Incorporating real UI elements** –
 - **Home Page / Hero Section:** Includes the actual project title, description, and “Upload Image” button.
 - **Upload Image Page:** Shows the uploaded image in real dimensions, with interactive upload functionality.
 - **Diagnosis Result Page:** Displays the predicted disease name, confidence score, and related plant information exactly as it would appear in the app.

High-fidelity wireframes for this project are essential for:

- **Presentations** to stakeholders, showing exactly how the system looks and works.
- **Usability testing**, allowing users to interact with a realistic version of the app.

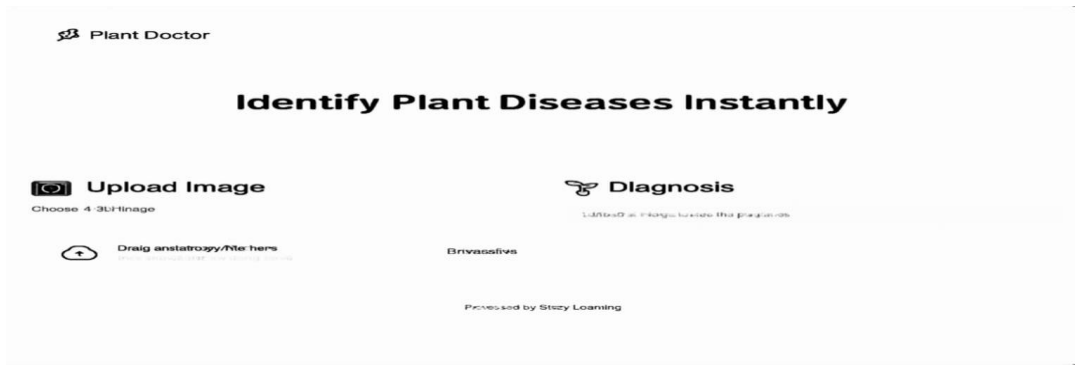


Figure 4.10 (High-Fidelity Wireframes- Homepage)

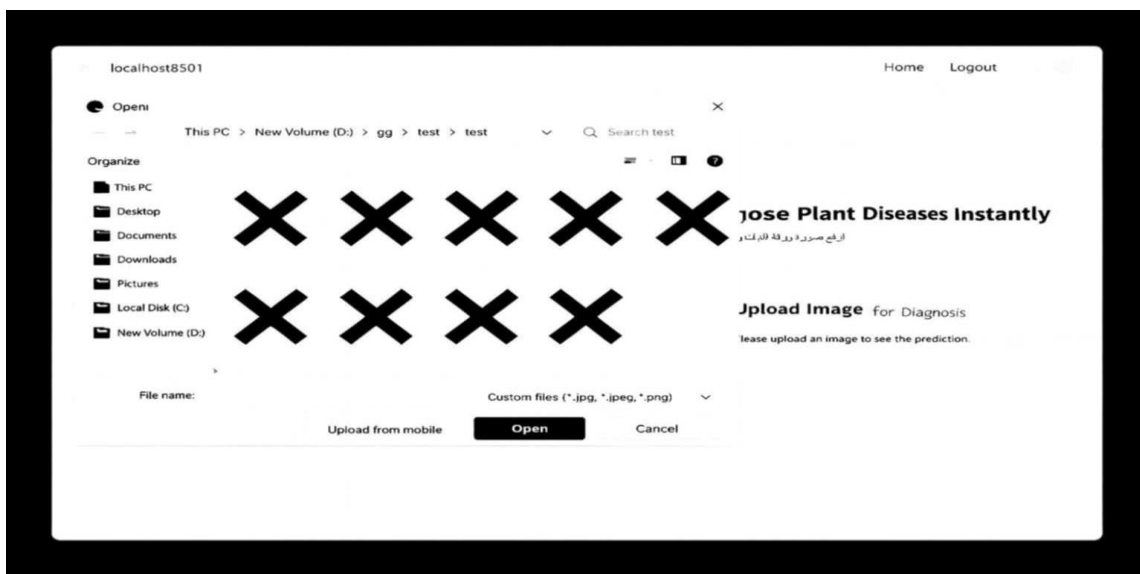


Figure 4.11 (High-Fidelity Wireframes- Upload Image page)



Figure 4.12 (High-Fidelity Wireframes- Diagnosis Result page)

4.5 User Interface (UI)

The User Interface (UI) is the visual part of the system that users interact with, including buttons, image upload areas, and result displays.

It is designed to be simple and clear so users can easily upload plant leaf images and view disease predictions. A clean UI improves usability and makes the system easier to use.

Steps to Design the UI in Figma for This Project:

- **Create a New Project** → A new Figma file is created to design the web-based interface of the Plant Disease Recognition System.
- **Add Screens (Frames)** → Web frames are added to represent the main screens of the system, such as the Home Page, Image Upload Screen, and Diagnosis Result Screen.
- **Insert Elements** → UI elements like the upload image button, image preview area, disease name text, confidence score, and navigation components are added based on the actual system workflow.
- **Set Colors and Fonts** → A consistent color scheme inspired by nature (green and white tones) is applied, along with clear and readable fonts to enhance accessibility and visual clarity.
- **Create Reusable Components** → Reusable components such as buttons and result cards are designed to maintain consistency across the interface and allow easy updates during development.



Figure 4.13 (Homepage UI)

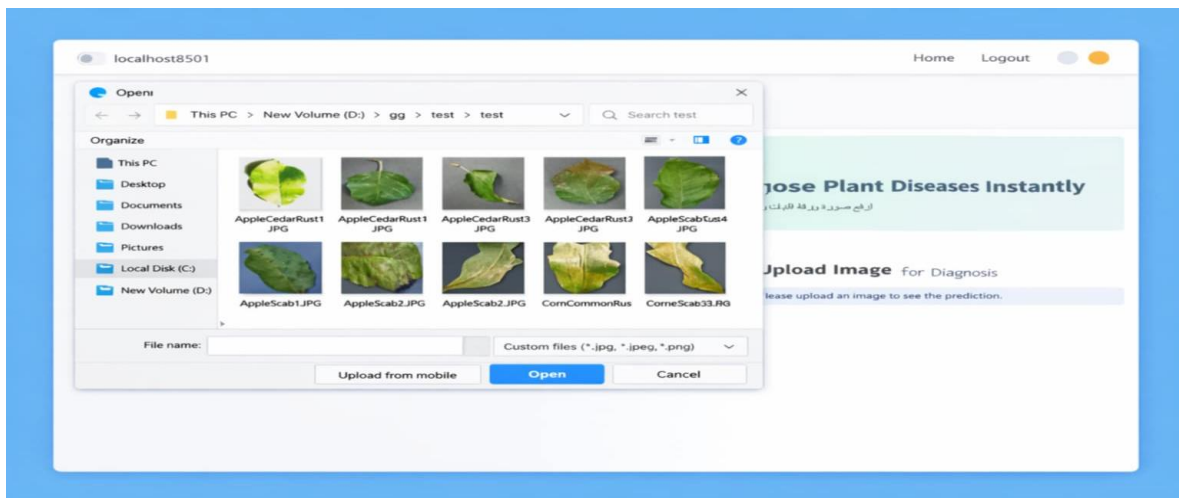


Figure 4.14 (Upload Image page UI)

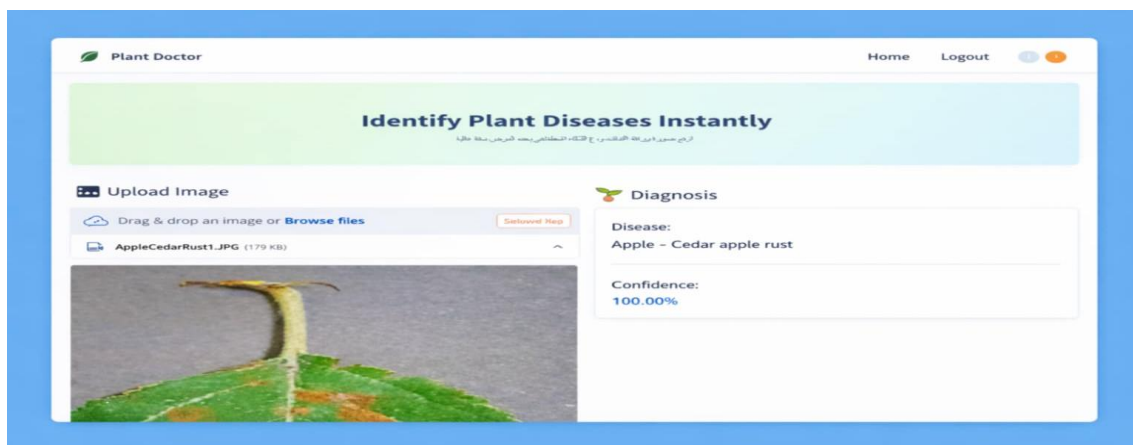


Figure 4.15 (Diagnosis Result page UI)

4.6 Design of Database / ERD

During the design of the Plant Disease Recognition System, it was determined that a database was not required. The system is designed to process uploaded plant leaf images and generate diagnosis results immediately using a CNN model. These results are displayed directly to the user and are not intended to be stored for future retrieval.

Since the system focuses on real-time disease Recognition rather than historical data analysis or user record management, introducing a database would add unnecessary complexity without providing significant value. Therefore, no database schema, Entity Relationship Diagram (ERD), or database mapping was implemented in this project.

4.7 Class Diagram of the Plant Disease Recognition System

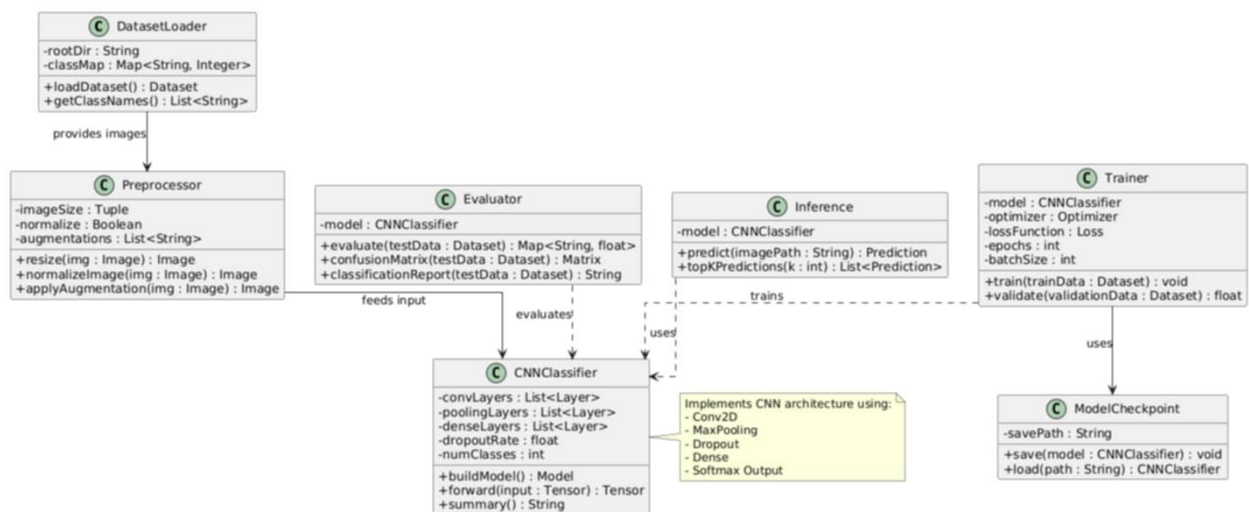


Figure 4.16(Class Diagram)

The diagram represents the overall structure and flow of the Plant Disease Recognition System for documentation purposes. It shows the main components and their interactions, including:

- 1. User Interface** – where users upload plant leaf images.
- 2. Image Preprocessing Module** – handles resizing, normalization, and preparation of images for the model.
- 3. CNN Model Module**– the core component that performs feature extraction and disease classification.
- 4. Result Modul** – generates and displays the diagnosis result with the disease name and confidence score.

4.8 Use-case Diagram of the Plant Disease Recognition System

The diagram illustrates the primary interactions between the user and the system. The main use-case represents the process of uploading a plant leaf image, which is then processed by the CNN model to generate a diagnosis.

Optional actions, such as retaking or re-uploading an image, are also included to allow users to correct mistakes or submit new samples. This diagram provides a clear conceptual overview of the system's functionality from the user's perspective, highlighting the key operations without detailing internal processing steps.

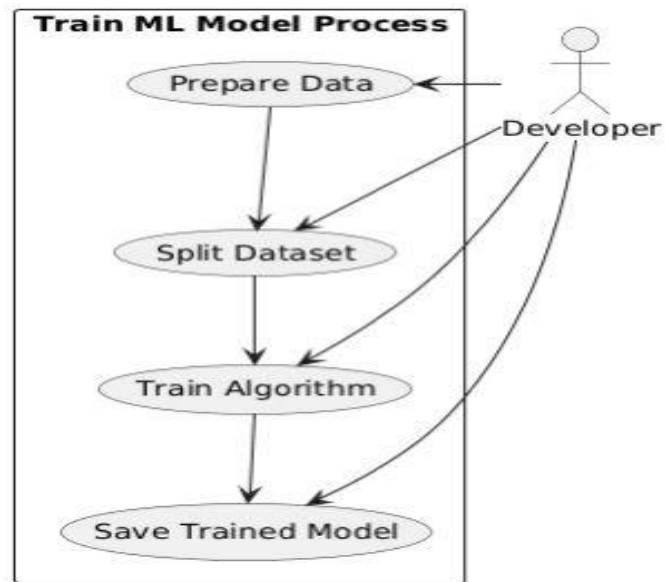


Figure 4.17(Use-case Diagram)

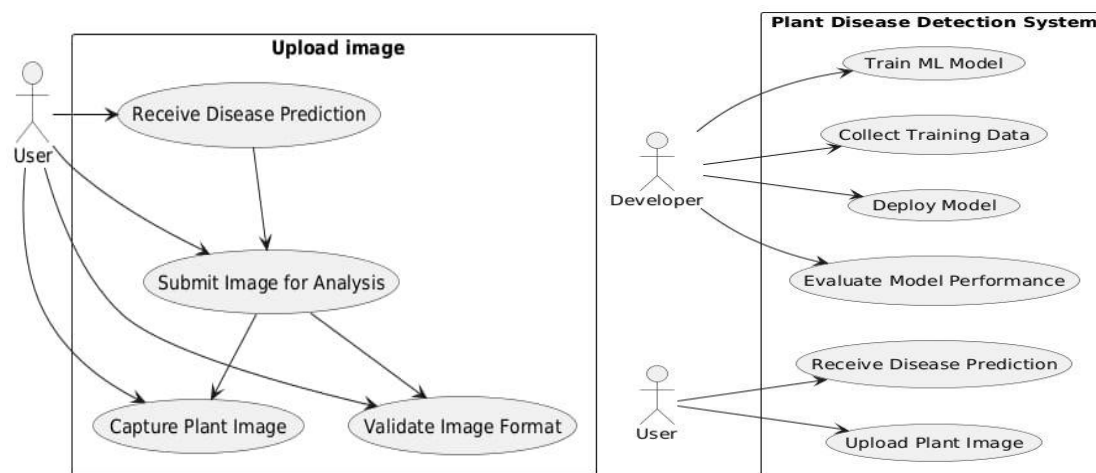


Figure 4.18(sub-Use-case Diagram)

4.9 Activity Diagram of the Plant Disease Recognition System

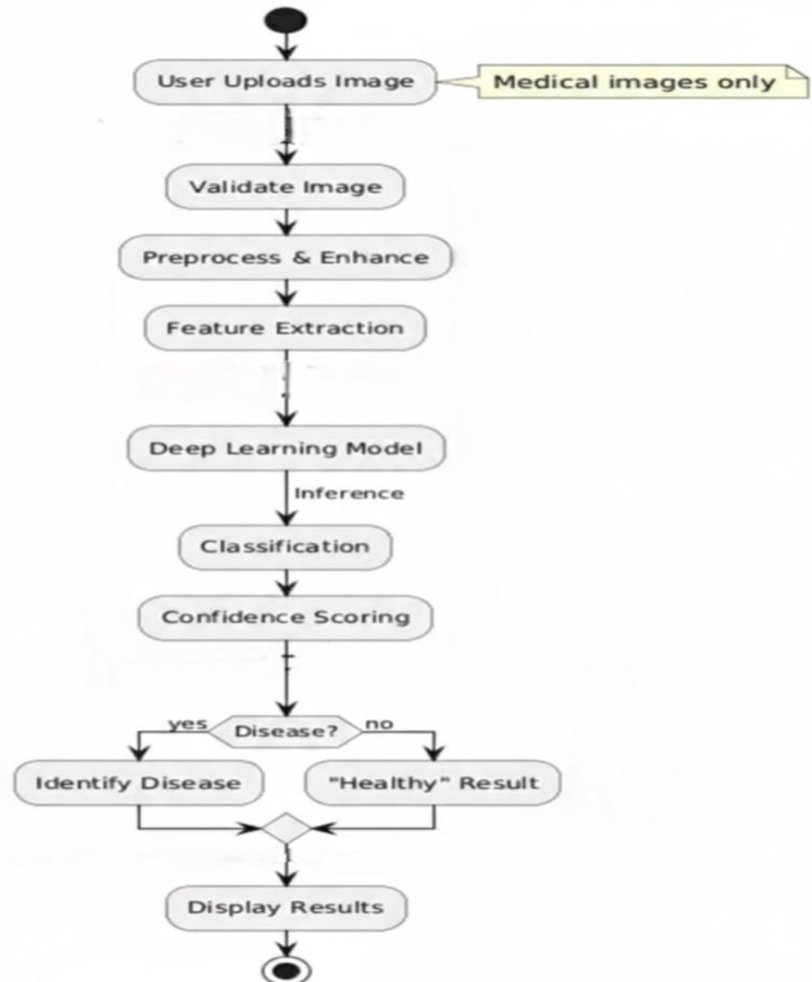


Figure 4.19(Activity Diagram)

The diagram illustrates the flow of operations within the system from the user's perspective. It begins with the user uploading a plant leaf image, followed by image preprocessing and processing through the CNN model for disease classification.

The system then generates and displays the diagnosis result, indicating whether the leaf is healthy or diseased, along with the confidence score. An optional step allows the user to retake or upload a new image if needed.

4.10 Sequence Diagram of the Plant Disease Recognition System

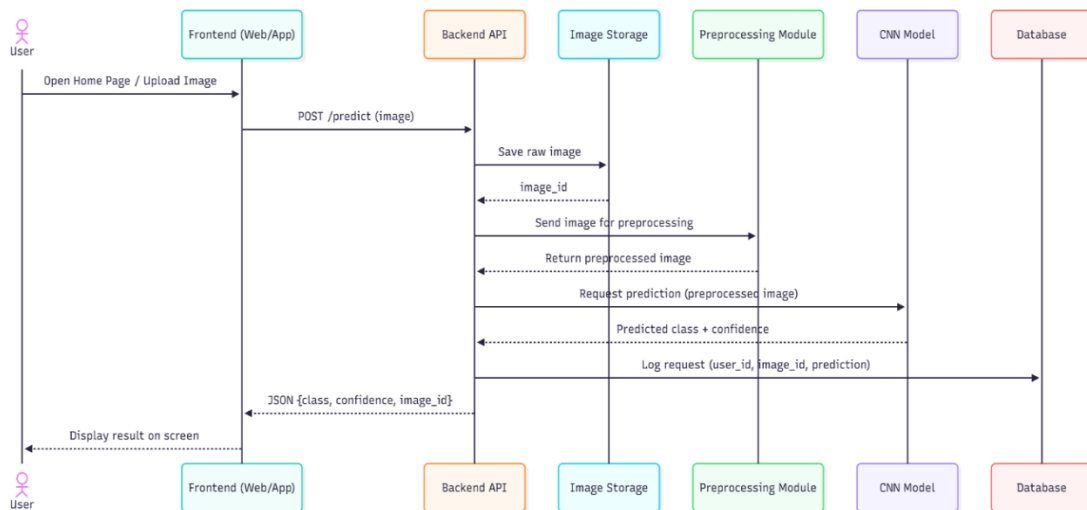


Figure 4.20(Sequence Diagram)

The sequence diagram illustrates the step-by-step interactions between the user and the system components. The process starts when the user uploads a plant leaf

4.11 Context Diagram of the Plant Disease Recognition System

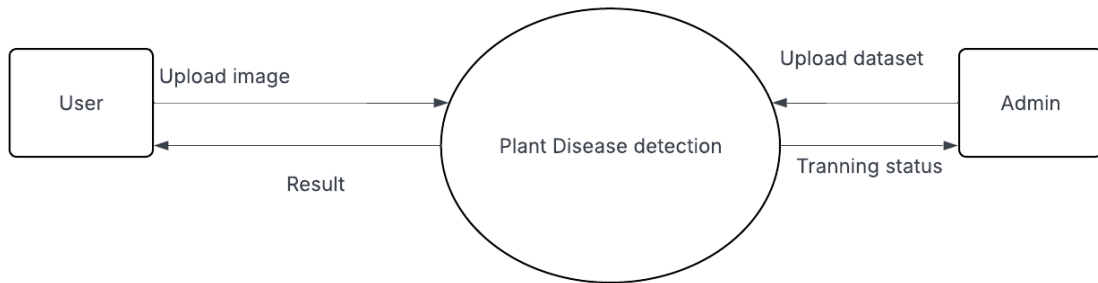


Figure 4.21(Context Diagram)

The figure illustrates a high-level system context diagram for a **Plant Disease Detection System**. The system is represented as a central processing entity that interacts with two primary external actors: the **User** and the **Admin**.

On the left side, the *User* interacts with the system by uploading an image of a plant leaf or crop. This image serves as the input data for analysis. After processing the image using the embedded detection model, the system returns the **result** to the user, which typically includes the identified disease classification and related diagnostic information.

Chapter 5

Implementation

5.1 Software Architecture

Software implementation is the process of developing and integrating all software components required for the Plant Disease Recognition System. The implementation includes model training, image preprocessing, prediction handling, and frontend integration.

Image Preprocessing

Image preprocessing is an essential step before prediction to prepare the input image for the CNN model. It includes resizing the image to 256×256 pixels, converting it into a NumPy array, normalizing pixel values by dividing by 255.0, and expanding its dimensions before feeding it into the model. These steps help ensure proper input formatting and improve prediction accuracy.

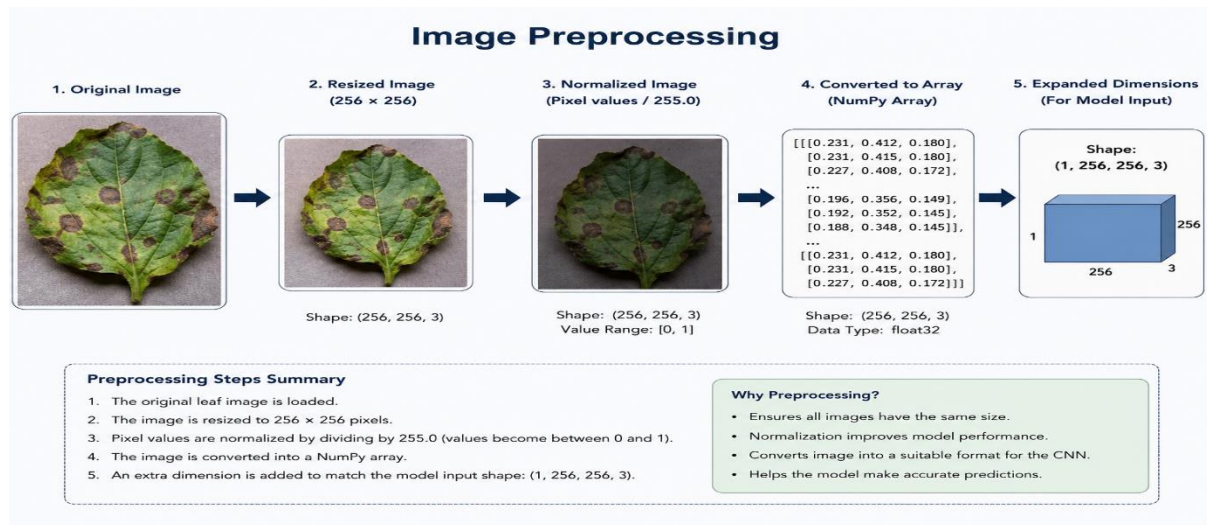


Figure 5.1: (Image Preprocessing)

CNN Architecture

The system uses a Convolutional Neural Network (CNN) for plant disease classification. The model extracts important visual features from leaf images, detects disease patterns such as spots and color changes, and classifies the image into the correct disease category.

The architecture consists of convolutional layers for feature extraction, pooling layers for reducing dimensions, and dense layers for final classification.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 256, 256, 32)	896
conv2d_1 (Conv2D)	(None, 256, 256, 32)	9,248
max_pooling2d (MaxPooling2D)	(None, 85, 85, 32)	0
conv2d_2 (Conv2D)	(None, 85, 85, 64)	18,496
conv2d_3 (Conv2D)	(None, 85, 85, 64)	36,928
max_pooling2d_1 (MaxPooling2D)	(None, 28, 28, 64)	0
conv2d_4 (Conv2D)	(None, 28, 28, 128)	73,856
conv2d_5 (Conv2D)	(None, 28, 28, 128)	147,584
max_pooling2d_2 (MaxPooling2D)	(None, 9, 9, 128)	0
conv2d_6 (Conv2D)	(None, 9, 9, 256)	295,168
conv2d_7 (Conv2D)	(None, 9, 9, 256)	590,080
conv2d_8 (Conv2D)	(None, 9, 9, 512)	3,277,312
conv2d_9 (Conv2D)	(None, 9, 9, 512)	6,554,112
flatten (Flatten)	(None, 41472)	0
dense (Dense)	(None, 1568)	65,029,664
dropout (Dropout)	(None, 1568)	0
dense_1 (Dense)	(None, 38)	59,622

Total params: 76,092,966 (290.27 MB)

Trainable params: 76,092,966 (290.27 MB)

Non-trainable params: 0 (0.00 B)

Figure 5.2: (CNN Architecture)

Disease Categories

The model supports multiple plant species and diseases, including:

- o Apple Diseases
- o Tomato Diseases
- o Potato Diseases
- o Corn Diseases
- o Grape Diseases
- o Pepper Diseases
- o Strawberry Diseases
- o Healthy Plant Detection

The system can classify healthy and diseased leaves across different crop categories.

5.2 User Interface

The frontend was developed using Streamlit to provide a simple and interactive interface for uploading plant leaf images and viewing prediction results. It displays the detected disease, confidence score, and recommendations, while offering a user-friendly design with image preview and clear result presentation.

System Visualization

This section presents the implementation of the Plant Disease Recognition System using a Streamlit-based interface. The figure illustrates the deployed application, including the process of loading the trained convolutional neural network model, preprocessing input images, and generating predictions in real time.

The system integrates the inference pipeline with a user-friendly interface, allowing users to upload plant leaf images and receive classification results along with confidence scores. In addition, the system retrieves relevant disease information, including descriptions, treatments, and prevention guidelines from an external knowledge base.

This implementation demonstrates the end-to-end workflow of the model from input acquisition to final prediction and recommendation generation.

```

1 import streamlit as st
2 from tensorflow.keras.models import load_model
3 from PIL import Image
4 import numpy as np
5 import json
6
7 # Load the model
8 st.set_page_config(
9     page_title="Plant Disease Recognition System",
10     page_icon="🌿",
11     layout="wide"
12 )
13
14 # Load the model
15 with open("D:\saved_models\trout.json", "r", encoding="utf-8") as f:
16     treatments = json.load(f)
17
18 # CSS
19 css = """
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
115
116
117
118
119
120
121
122
123
124
125
126
127
128
129
130
131
132
133
134
135
136
137
138
139
140
141
142
143
144
145
146
147
148
149
150
151
152
153
154
155
156
157
158
159
160
161
162
163
164
165
166
167
168
169
170
171
172
173
174
175
176
177
178
179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
217
218
219
220
221
222
223
224
225
226
227
228
229
230
231
232
233
234
235
236
237
238
239
240
241
242
243
244
245
246
247
248
249
250
251
252
253
254
255
256
257
258
259
260
261
262
263
264
265
266
267
268
269
270
271
272
273
274
275
276
277
278
279
280
281
282
283
284
285
286
287
288
289
290
291
292
293
294
295
296
297
298
299
300
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
327
328
329
330
331
332
333
334
335
336
337
338
339
340
341
342
343
344
345
346
347
348
349
350
351
352
353
354
355
356
357
358
359
360
361
362
363
364
365
366
367
368
369
370
371
372
373
374
375
376
377
378
379
380
381
382
383
384
385
386
387
388
389
390
391
392
393
394
395
396
397
398
399
400
401
402
403
404
405
406
407
408
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432
433
434
435
436
437
438
439
440
441
442
443
444
445
446
447
448
449
450
451
452
453
454
455
456
457
458
459
460
461
462
463
464
465
466
467
468
469
470
471
472
473
474
475
476
477
478
479
480
481
482
483
484
485
486
487
488
489
490
491
492
493
494
495
496
497
498
499
500
501
502
503
504
505
506
507
508
509
510
511
512
513
514
515
516
517
518
519
520
521
522
523
524
525
526
527
528
529
530
531
532
533
534
535
536
537
538
539
540
541
542
543
544
545
546
547
548
549
550
551
552
553
554
555
556
557
558
559
560
561
562
563
564
565
566
567
568
569
570
571
572
573
574
575
576
577
578
579
580
581
582
583
584
585
586
587
588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647
648
649
650
651
652
653
654
655
656
657
658
659
660
661
662
663
664
665
666
667
668
669
670
671
672
673
674
675
676
677
678
679
680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755
756
757
758
759
760
761
762
763
764
765
766
767
768
769
770
771
772
773
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809
810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863
864
865
866
867
868
869
870
871
872
873
874
875
876
877
878
879
880
881
882
883
884
885
886
887
888
889
890
891
892
893
894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917
918
919
920
921
922
923
924
925
926
927
928
929
930
931
932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971
972
973
974
975
976
977
978
979
980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013
1014
1015
1016
1017
1018
1019
1020
1021
1022
1023
1024
1025
1026
1027
1028
1029
1030
1031
1032
1033
1034
1035
1036
1037
1038
1039
1040
1041
1042
1043
1044
1045
1046
1047
1048
1049
1050
1051
1052
1053
1054
1055
1056
1057
1058
1059
1060
1061
1062
1063
1064
1065
1066
1067
1068
1069
1070
1071
1072
1073
1074
1075
1076
1077
1078
1079
1080
1081
1082
1083
1084
1085
1086
1087
1088
1089
1090
1091
1092
1093
1094
1095
1096
1097
1098
1099
1100
1101
1102
1103
1104
1105
1106
1107
1108
1109
1110
1111
1112
1113
1114
1115
1116
1117
1118
1119
1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131
1132
1133
1134
1135
1136
1137
1138
1139
1140
1141
1142
1143
1144
1145
1146
1147
1148
1149
1150
1151
1152
1153
1154
1155
1156
1157
1158
1159
1160
1161
1162
1163
1164
1165
1166
1167
1168
1169
1170
1171
1172
1173
1174
1175
1176
1177
1178
1179
1180
1181
1182
1183
1184
1185
1186
1187
1188
1189
1190
1191
1192
1193
1194
1195
1196
1197
1198
1199
1200
1201
1202
1203
1204
1205
1206
1207
1208
1209
1210
1211
1212
1213
1214
1215
1216
1217
1218
1219
1220
1221
1222
1223
1224
1225
1226
1227
1228
1229
1230
1231
1232
1233
1234
1235
1236
1237
1238
1239
1240
1241
1242
1243
1244
1245
1246
1247
1248
1249
1250
1251
1252
1253
1254
1255
1256
1257
1258
1259
1260
1261
1262
1263
1264
1265
1266
1267
1268
1269
1270
1271
1272
1273
1274
1275
1276
1277
1278
1279
1280
1281
1282
1283
1284
1285
1286
1287
1288
1289
1290
1291
1292
1293
1294
1295
1296
1297
1298
1299
1300
1301
1302
1303
1304
1305
1306
1307
1308
1309
1310
1311
1312
1313
1314
1315
1316
1317
1318
1319
1320
1321
1322
1323
1324
1325
1326
1327
1328
1329
1330
1331
1332
1333
1334
1335
1336
1337
1338
1339
1340
1341
1342
1343
1344
1345
1346
1347
1348
1349
1350
1351
1352
1353
1354
1355
1356
1357
1358
1359
1360
1361
1362
1363
1364
1365
1366
1367
1368
1369
1370
1371
1372
1373
1374
1375
1376
1377
1378
1379
1380
1381
1382
1383
1384
1385
1386
1387
1388
1389
1390
1391
1392
1393
1394
1395
1396
1397
1398
1399
1400
1401
1402
1403
1404
1405
1406
1407
1408
1409
1410
1411
1412
1413
1414
1415
1416
1417
1418
1419
1420
1421
1422
1423
1424
1425
1426
1427
1428
1429
1430
1431
1432
1433
1434
1435
1436
1437
1438
1439
1440
1441
1442
1443
1444
1445
1446
1447
1448
1449
1450
1451
1452
1453
1454
1455
1456
1457
1458
1459
1460
1461
1462
1463
1464
1465
1466
1467
1468
1469
1470
1471
1472
1473
1474
1475
1476
1477
1478
1479
1480
1481
1482
1483
1484
1485
1486
1487
1488
1489
1490
1491
1492
1493
1494
1495
1496
1497
1498
1499
1500
1501
1502
1503
1504
1505
1506
1507
1508
1509
1510
1511
1512
1513
1514
1515
1516
1517
1518
1519
1520
1521
1522
1523
1524
1525
1526
1527
1528
1529
1530
1531
1532
1533
1534
1535
1536
1537
1538
1539
1540
1541
1542
1543
1544
1545
1546
1547
1548
1549
1550
1551
1552
1553
1554
1555
1556
1557
1558
1559
1560
1561
1562
1563
1564
1565
1566
1567
1568
1569
1570
1571
1572
1573
1574
1575
1576
1577
1578
1579
1580
1581
1582
1583
1584
1585
1586
1587
1588
1589
1590
1591
1592
1593
1594
1595
1596
1597
1598
1599
1600
1601
1602
1603
1604
1605
1606
1607
1608
1609
1610
1611
1612
1613
1614
1615
1616
1617
1618
1619
1620
1621
1622
1623
1624
1625
1626
1627
1628
1629
1630
1631
1632
1633
1634
1635
1636
1637
1638
1639
1640
1641
1642
1643
1644
1645
1646
1647
1648
1649
1650
1651
1652
1653
1654
1655
1656
1657
1658
1659
1660
1661
1662
1663
1664
1665
1666
1667
1668
1669
1670
1671
1672
1673
1674
1675
1676
1677
1678
1679
1680
1681
1682
1683
1684
1685
1686
1687
1688
1689
1690
1691
1692
1693
1694
1695
1696
1697
1698
1699
1700
1701
1702
1703
1704
1705
1706
1707
1708
1709
1710
1711
1712
1713
1714
1715
1716
1717
1718
1719
1720
1721
1722
1723
1724
1725
1726
1727
1728
1729
1730
1731
1732
1733
1734
1735
1736
1737
1738
1739
1740
1741
1742
1743
1744
1745
1746
1747
1748
1749
1750
1751
1752
1753
1754
1755
1756
1757
1758
1759
1760
1761
1762
1763
1764
1765
1766
1767
1768
1769
1770
1771
1772
1773
1774
1775
1776
1777
1778
1779
1780
1781
1782
1783
1784
1785
1786
1787
1788
1789
1790
1791
1792
1793
1794
1795
1796
1797
1798
1799
1800
1801
1802
1803
1804
1805
1806
1807
1808
1809
1810
1811
1812
1813
1814
1815
1816
1817
1818
1819
1820
1821
1822
1823
1824
1825
1826
1827
1828
1829
1830
1831
1832
1833
1834
1835
1836
1837
1838
1839
1840
1841
1842
1843
1844
1845
1846
1847
1848
1849
1850
1851
1852
1853
1854
1855
1856
1857
1858
1859
1860
1861
1862
1863
1864
1865
1866
1867
1868
1869
1870
1871
1872
1873
1874
1875
1876
1877
1878
1879
1880
1881
1882
1883
1884
1885
1886
1887
1888
1889
1890
1891
1892
1893
1894
1895
1896
1897
1898
1899
1900
1901
1902
1903
1904
1905
1906
1907
1908
1909
1910
1911
1912
1913
1914
1915
1916
1917
1918
1919
1920
1921
1922
1923
1924
1925
1926
1927
1928
1929
1930
1931
1932
1933
1934
1935
1936
1937
1938
1939
1940
1941
1942
1943
1944
1945
1946
1947
1948
1949
1950
1951
1952
1953
1954
1955
1956
1957
1958
1959
1960
1961
1962
1963
1964
1965
1966
1967
1968
1969
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
2036
2037
2038
2039
2040
2041
2042
2043
2044
2045
2046
2047
2048
2049
2050
2051
2052
2053
2054
2055
2056
2057
2058
2059
2060
2061
2062
2063
2064
2065
2066
2067
2068
2069
2070
2071
2072
2073
2074
2075
2076
2077
2078
2079
2080
2081
2082
2083
2084
2085
2086
2087
2088
2089
2090
2091
2092
2093
2094
2095
2096
2097
2098
2099
2100
2101
2102
2103
2104
2105
2106
2107
2108
2109
2110
2111
2112
2113
2114
2115
2116
2117
2118
2119
2120
2121
2122
2123
2124
2125
2126
2127
2128
2129
2130
2131
2132
2133
2134
2135
2136
2137
2138
2139
2140
2141
2142
2143
2144
2145
2146
2147
2148
2149
2150
2151
2152
2153
2154
2155
2156
2157
2158
2159
2160
2161
2162
2163
2164
2165
2166
2167
2168
2169
2170
2171
2172
2173
2174
2175
2176
2177
2178
2179
2180
2181
2182
2183
2184
2185
2186
2187
2188
2189
2190
2191
2192
2193
2194
2195
2196
2197
2198
2199
2200
2201
2202
2203
2204
2205
2206
2207
2208
2209
2210
2211
2212
2213
2214
2215
2216
2217
2218
2219
2220
2221
2222
2223
2224
2225
2226
2227
2228
2229
2230
2231
2232
2233
2234
2235
2236
2237
2238
2239
2240
2241
2242
2243
2244
2245
2246
2247
2248
2249
2250
2251
2252
2253
2254
2255
2256
2257
2258
2259
2260
2261
2262
2263
2264
2265
2266
2267
2268
2269
2270
2271
2272
2273
2274
2275
2276
2277
2278
2279
2280
2281
2282
2283
2284
2285
2286
2287
2288
2289
2290
2291
2292
2293
2294
2295
2296
2297
2298
2299
2300
2301
2302
2303
2304
2305
2306
2307
2308
2309
2310
2311
2312
2313
2314
2315
2316
2317
2318
2319
2320
2321
2322
2323
2324
2325
2326
2327
2328
2329
2330
2331
2332
2333
2334
2335
2336
2337
2338
2339
2340
2341
2342
2343
2344
2345
2346
2347
2348
2349
2350
2351
2352
2353
2354
2355
2356
2357
2358
2359
2360
2361
2362
2363
2364
2365
2366
2367
2368
2369
2370
2371
2372
2373
2374
2375
2376
2377
2378
2379
2380
2381
2382
2383
2384
2385
2386
2387
2388
2389
2390
2391
2392
2393
2394
2395
2396
2397
2398
2399
2400
2401
2402
2403
2404
2405
2406
2407
2408
2409
2410
2411
2412
2413
2414
2415
2416
2417
2418
2419
2420
2421
2422
2423
2424
2425
2426
2427
2428
2429
2430
2431
2432
2433
2434
2435
2436
2437
2438
2439
2440
2441
2442
2443
2444
2445
2446
2447
2448
2449
2450
2451
2452
2453
2454
2455
2456
2457
2458
2459
2460
2461
2462
2463
2464
2465
2466
2467
2468
2469
2470
2471
2472
2473
2474
2475
2476
2477
2478
2479
2480
2481
2482
2483
2484
2485
2486
2487
2488
2489
2490
2491
2492
2493
2494
2495
2496
2497
2498
2499
2500
2501
2502
2503
2504
2505
2506
2507
2508
2509
2510
2511
2512
2513
2514
2515
2516
2517
2518
2519
2520
2521
2522
2523
2524
2525
2526
2527
2528
2529
2530
2531
2532
2533
2534
2535
2536
2537
2538
2539
2540
2541
2542
2543
2544
2545
2546
2547
2548
2549
2550
2551
2552
2553
2554
2555
2556
2557
2558
2559
2560
2561
2562
2563
2564
2565
2566
2567
2568
2569
2570
2571
2572
2573
2574
2575
2576
2577
2578
2579
2580
2581
2582
2583
2584
2585
2586
2587
2588
2589
2590
2591
2592
2593
2594
2595
2596
2597
2598
2599
2600
2601
2602
2603
2604
2605
2606
2607
2608
2609
2610
2611
2612
2613
2614
2615
2616
2617
2618
2619
2620
2621
2622
2623
2624
2625
2626
2627
2628
2629
2630
2631
2632
2633
2634
2635
2636
2637
2638
2639
2640
2641
2642
2643
2644
2645
2646
2647
2648
2649
2650
2651
2652
2653
2654
2655
2656
2657
2658
2659
2660
2661
2662
2663
2664
2665
2666
2667
2668
2669
2670
2671
2672
2673
267
```

5.3 Results and Discussion

The performance of the Plant Disease Recognition System was evaluated using different plant leaf images from multiple disease categories. The trained CNN model demonstrated strong classification capabilities and achieved high prediction accuracy during testing.

The training and validation accuracy curves indicate that the model learned the disease patterns effectively throughout the training process. At the same time, the loss curves showed a continuous decrease, which reflects the stability of the learning process and the ability of the model to generalize on unseen data.

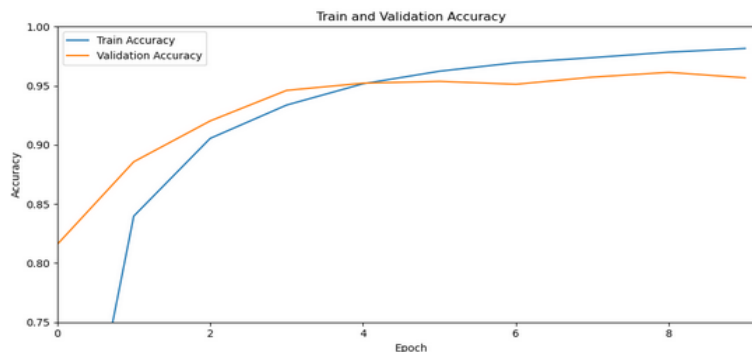


Figure 5.4: (Training and Validation Accuracy)

The accuracy graph shows that both training and validation accuracy increased gradually during training. The small gap between the two curves indicates that the model achieved good generalization performance and did not suffer significantly from overfitting.

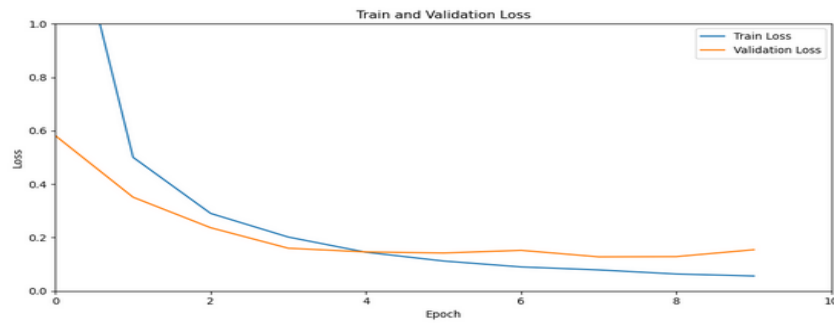


Figure 5.5: (Training and Validation Loss)

The loss graph demonstrates a consistent reduction in both training and validation loss values. This behavior confirms that the model successfully optimized its parameters and improved prediction performance over successive epochs.

The developed system was further tested using different plant leaf images through the Streamlit interface. The system successfully identified plant diseases and provided confidence scores along with treatment and prevention recommendations.

Treatment Recommendation System

The system includes a recommendation feature that provides users with disease-related information after prediction.

The recommendation system retrieves data from a JSON file containing:

- o Disease description.
- o Treatment methods.
- o Prevention recommendations..

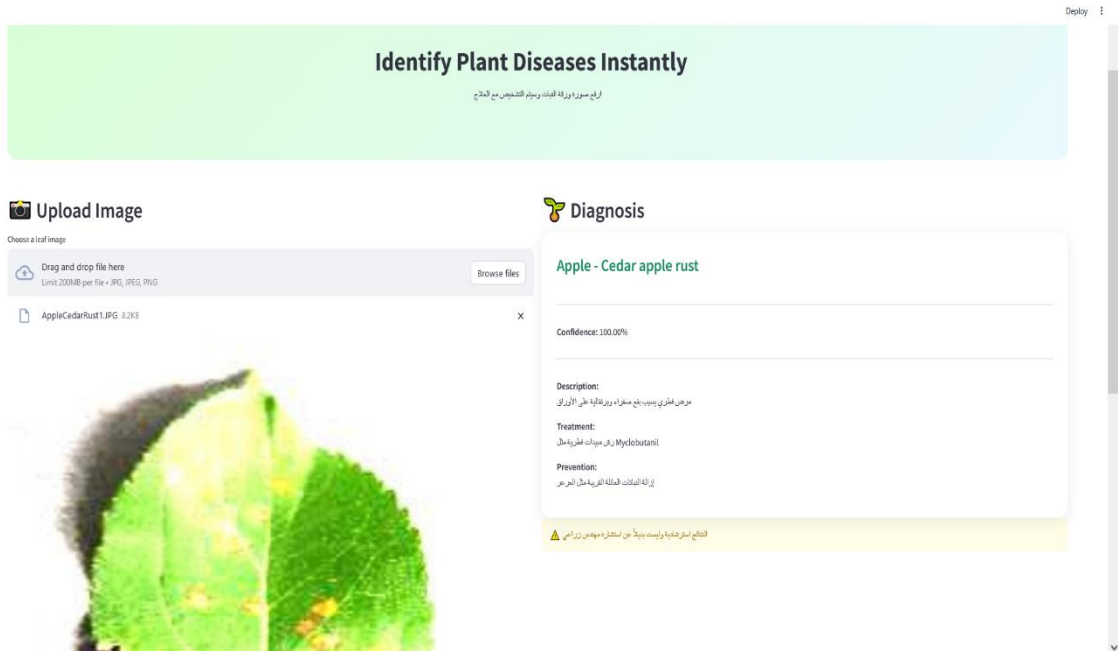


Figure 5.6: (Treatment Recommendation System)

System Features

Key Features of the System

- Automatic detection of plant diseases using a CNN-based deep learning model.
- Fast and accurate prediction with confidence scores.
- User-friendly interface for image upload and result display.
- Disease descriptions, treatment suggestions, and prevention recommendations.
- Integration of TensorFlow/Keras with Streamlit for efficient disease classification.

Chapter 6

Testing

6.1 Unit Testing

Unit testing was conducted to verify that each module of the Plant Disease Recognition System functions correctly.

Module	Expected Result	Status
Image Upload	Image uploaded successfully	Pass
Image Preprocessing	Image resized and normalized correctly	Pass
CNN Prediction	Disease predicted correctly	Pass
Confidence Score	Confidence displayed successfully	Pass
Treatment Recommendation	Treatment information retrieved correctly	Pass
User Interface	Results displayed successfully	Pass

The testing results indicate that all individual modules operated successfully and produced the expected outputs.

6.2 Integration Testing

Integration testing was performed to ensure that all system components work together correctly.

The integration workflow consists of:

1. Uploading a plant leaf image.
2. Preprocessing the image.
3. Predicting the disease using the CNN model.
4. Retrieving treatment information.
5. Displaying the final results.

Overall System Architecture of the Plant Disease Recognition System

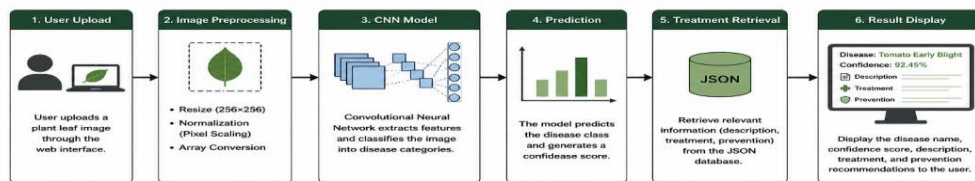


Figure 6.1: (System Integration Workflow)

Scenario	Expected Result	Status
Diseased Leaf Image	Correct disease prediction	Pass
Healthy Leaf Image	Healthy class displayed	Pass
Treatment Retrieval	Treatment information displayed	Pass
Final Output Display	Complete results displayed successfully	Pass

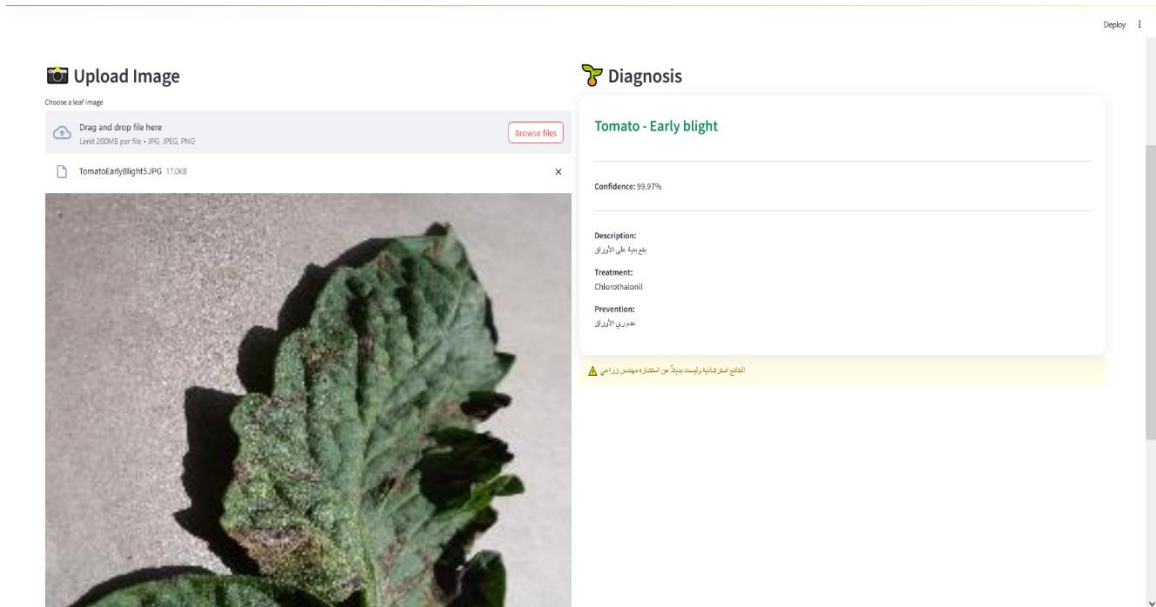


Figure 6.2: (Integrated System Output)

The integration testing results confirm that all modules communicate effectively and operate as a complete system. The application successfully processes plant leaf images, predicts diseases, retrieves recommendations, and displays the results through the user interface.

Conclusion

This project presented the design and implementation of a Plant Disease Recognition System based on deep learning techniques. The primary objective of the system was to assist users in identifying plant diseases automatically through image analysis and to provide useful recommendations for treatment and prevention.

A Convolutional Neural Network (CNN) model was developed and trained using a dataset containing images of healthy and diseased plant leaves. The model learned to extract important visual features from the images and classify them into the appropriate disease categories. Image preprocessing techniques, including resizing and normalization, were applied to improve model performance and ensure consistent input data.

To make the system practical and accessible, a user-friendly web application was developed using Streamlit. The application enables users to upload plant leaf images and receive instant predictions along with confidence scores. In addition, the system provides disease descriptions, treatment methods, and prevention guidelines through an integrated recommendation module. This feature enhances the usefulness of the application by offering not only diagnosis but also guidance for disease management.

The implementation and testing phases demonstrated that the developed system operates successfully and achieves its intended objectives. Unit testing verified the functionality of individual modules, while integration testing confirmed the proper interaction between all system components. The results showed that the model is capable of accurately recognizing different plant diseases and distinguishing between healthy and infected leaves.

The project highlights the potential of artificial intelligence and deep learning in supporting modern agriculture. By automating disease detection, the system can help farmers, agricultural specialists, and plant owners identify diseases more quickly and take appropriate actions at an early stage. Early detection can reduce crop losses, improve plant health, and contribute to more efficient agricultural practices.

In conclusion, the Plant Disease Recognition System successfully combines deep learning technology with an interactive web interface to provide an effective and reliable solution for plant disease diagnosis. Future improvements may include expanding the dataset, supporting additional plant species and diseases, enhancing model accuracy, and deploying the system on mobile platforms to increase accessibility and real-world usability.

Future Work

The Plant Disease Recognition System represents an effective application of deep learning techniques for identifying plant diseases using image classification. Although the current system achieves high accuracy and provides reliable predictions, there are several possible improvements and enhancements that can be implemented in future versions to increase its performance, scalability, and usability.

One of the main future improvements is the expansion of the dataset to include a larger variety of plant species and disease categories. Increasing the dataset size and diversity would help improve the model's generalization ability and make it more robust when tested on real-world agricultural environments.

Another important enhancement is improving the deep learning architecture by experimenting with more advanced models such as transfer learning using pre-trained networks (e.g., ResNet, EfficientNet, or MobileNet). These models may provide higher accuracy and faster convergence compared to the current CNN architecture.

In addition, the system can be improved by integrating real-time disease detection using mobile devices. A mobile application version of the system would allow farmers and users to capture plant images directly from the field and receive instant predictions and recommendations.

Future work may also include deploying the system on cloud platforms to make it accessible to a larger number of users. Cloud deployment would allow better scalability, faster processing, and centralized model updates without requiring users to install the system locally.

Furthermore, the recommendation system can be enhanced by incorporating more detailed agricultural knowledge, including pesticide suggestions, severity levels of diseases, and localized treatment options based on geographic regions.

Another potential improvement is adding explainable AI (XAI) techniques to provide better interpretability of model predictions. This would help users understand the reasons behind each prediction, increasing trust in the system.

Finally, the user interface can be further improved by making it more interactive and responsive, with support for multiple languages and better visualization of prediction results.

In conclusion, the system has strong potential for future development, and with the suggested improvements, it can become a more powerful and practical tool for modern agriculture and plant disease management.

References

1. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
Available at: <https://www.deeplearningbook.org/>
2. Chollet, F. (2017). *Deep Learning with Python*. Manning Publications.
3. TensorFlow Documentation. (2024). *TensorFlow Guide*.
Available at: <https://www.tensorflow.org/>
4. Keras Documentation. (2024). *Keras API Reference*.
Available at: <https://keras.io/>
5. Streamlit Documentation. (2024). *Streamlit: Build Web Apps for Machine Learning*.
Available at: <https://docs.streamlit.io/>
6. PlantVillage Dataset. (2023). *Plant Disease Image Dataset*.
Available at: <https://www.kaggle.com/datasets/emmarex/plantdisease>
7. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). *Using Deep Learning for Image-Based Plant Disease Detection*. *Frontiers in Plant Science*.
8. OpenCV Documentation. (2024). *OpenCV Image Processing Library*.
Available at: <https://opencv.org/>
9. Python Software Foundation. (2024). *Python Language Reference*.
Available at: <https://www.python.org/>
10. Chollet, F. (2015). *Convolutional Neural Networks for Image Classification*. (Concepts used in CNN architectures).

Appendices: GitHub Link

- <https://github.com/Zn634677/Agri-AI-Intelligent-Plant-Disease-Recognition-System-deep-learning>

نظام ذكي لاكتشاف وتشخيص أمراض النباتات بالاعتماد على تقنيات التعلم العميق

يهدف المشروع الي تطوير نظام ذكي للتعرف على أمراض النباتات باستخدام تقنيات التعلم العميق، وذلك للمساهمة في تحسين الإنتاج الزراعي وتقليل الخسائر الناتجة عن الأمراض النباتية. تمثل أمراض النباتات أحد أهم التحديات التي تواجه القطاع الزراعي، حيث يؤدي التأخر في اكتشافها إلى انخفاض الإنتاجية وزيادة التكاليف الاقتصادية.

- Convolutional Neural Networks) يعتمد النظام المقترح على الشبكات العصبية التلافيفية لتحليل صور أوراق النباتات وتصنيفها إلى نباتات سليمة أو مصابة بأمراض مختلفة بدقة (CNN) التي تحتوي على آلاف الصور المصنفة Plant Village عالية. وقد تم استخدام مجموعة بيانات مسبقاً، مع تطبيق عدد من تقنيات المعالجة المسبقة للصور مثل تغيير الحجم، وتطبيع البيانات، لتحسين أداء النموذج وتقليل مشكلة الإفراط في (Data Augmentation) وتعزيز البيانات (Overfitting). التخصيص.

والاعتماد على مجموعة من الأدوات والمكتبات Python تم تطوير النظام باستخدام لغة البرمجة يتكون النظام من أربع وحدات رئيسية. OpenCV و Keras و TensorFlow المتخصصة، أبرزها ، ووحدة CNN تشمل واجهة المستخدم، ووحدة معالجة الصور، ونموذج التصنيف القائم على عرض النتائج. يتيح النظام للمستخدم رفع صورة لورقة النبات ليقوم بتحليلها وإظهار التشخيص المناسب مع درجة الثقة في النتيجة خلال وقت قصير.

أظهرت النتائج التجريبية كفاءة النموذج في التعرف على أمراض النباتات بدقة مرتفعة، مما يؤكد فعالية استخدام تقنيات الذكاء الاصطناعي والتعلم العميق في المجال الزراعي. ويساهم النظام في دعم المزارعين والمهندسين الزراعيين من خلال توفير أداة سريعة وسهلة الاستخدام للكشف المبكر عن الأمراض، الأمر الذي يساعد على اتخاذ الإجراءات العلاجية المناسبة وتقليل الاعتماد غير الضروري على المبيدات. كما يوصى مستقبلاً بتوسيع قاعدة البيانات المستخدمة وتطوير النظام ليعمل بكفاءة أكبر في البيئات الحقلية الحقيقية ذات ظروف الإضاءة والخلفيات المتغيرة.